Anytime-valid weighted survival tests for exact failures and grouped assessments

Hien Duy Nguyen[1,2] and Liming Xiang[3]

[1]School of Computing, Engineering and Mathematical Sciences,
La Trobe University, Bundoora, Victoria, Australia

[2]Institute of Mathematics for Industry,
Kyushu University, Fukuoka, Japan

[3]School of Physical and Mathematical Sciences,
Nanyang Technological University, Singapore

Abstract

Repeated application of a fixed-analysis weighted logrank test does not retain its nominal type-I error guarantee. Within the conditional likelihood-ratio framework of ter Schure et al. (2024), we use classical survival weights to specify time-varying alternatives. The construction covers exact failures with right censoring and first detections within common completed assessment windows. Its finite-sample guarantee requires conditional exchangeability of event allocation within each comparison stratum. Multiplying the conditional likelihood ratios then gives a test martingale, including when valid exact and grouped comparisons are combined. The derivative of its log at the null is the corresponding weighted Cochran–Mantel–Haenszel or logrank score. A fixed-margin calculation relates relative weights to local score shifts; it does not optimize finite-duration rejection probability. Simulations show that weight comparisons depend on coefficient choice: a delayed-effect advantage of Fleming–Harrington weighting persists under fixed specifications, whereas an early-effect advantage of Peto–Peto weighting does not. Monitoring flexibility can entail substantial power loss at small sample sizes. Retrospective clinical examples illustrate the additional assumptions and information loss introduced by grouping irregular visits.

Keywords: e-values; right censoring; interval censoring; sequential monitoring; weighted logrank tests; covariate stratification.

1 Introduction

Survival comparisons involve two design choices: which differences over follow-up matter, and when the evidence will be examined. The classical logrank test is effective under approximately proportional hazards (Mantel, 1966; Cox, 1972). Weighted tests give more emphasis to particular stages of follow-up: Gehan–Breslow weights emphasize earlier failures (Gehan, 1965), Fleming–Harrington weights can target early or delayed differences (Harrington and Fleming, 1982), and Gray–Tsiatis weighting is motivated by long-term survival differences (Gray and Tsiatis, 1989). Repeatedly applying a fixed-analysis test and stopping at significance can inflate the false-positive probability. Group-sequential boundaries and alpha-spending methods address planned monitoring (O'Brien and Fleming, 1979; Lan and DeMets, 1983; Slud, 1984). An anytime-valid test instead controls the probability of ever crossing its rejection threshold, including under an unplanned extension of follow-up.

The observation scheme determines which comparisons support that guarantee. Exact failures with right censoring lead to risk-set comparisons (Andersen et al., 1993). Interval assessments, such as angiography for cardiac allograft vasculopathy, locate the event between a last negative and a first positive visit (Sharples et al., 2003). Imputing an exact failure time changes the ordering used by a rank test (Law and Brookmeyer, 1992). Existing interval-censoring methods use the intervals directly for survival estimation (Turnbull, 1976), regression (Finkelstein, 1986), and group comparison (Sun, 1996; Fay and Shaw, 2010); see Sun (2006). These fixed-analysis methods do not by themselves justify unrestricted repeated testing. Here we study first detections in common completed assessment windows, rather than arbitrary individual interval-censoring schedules.

The validity mechanism is inherited from ter Schure et al. (2024). Their equation (3.5) permits past-dependent numerator distributions. For grouped assessments, their Section 3.3 uses the same hypergeometric conditional null as in this work. A normalized working alternative divided by that null gives a factor with conditional mean one; multiplying the factors yields a test martingale. Our weighted exponential tilts are a particular family of admissible numerators within this construction, not a new martingale principle. Stratified count-table e-values are also developed by Turner and Grünwald (2023); survival data additionally require attention to changing risk sets, absorbing failures and incomplete observation.

We use classical survival weights to specify working alternatives of the form

$$q^{(w)}_{\eta,j}(y) \;\propto\; p_{0j}(y)\exp(\eta w_j y),$$

where p_{0j} is the conditional null mass, w_j is a weight and η determines the direction and magnitude of the tilt. The choices cannot use the current group allocation of events. We identify the weighted Cochran–Mantel–Haenszel (CMH) or logrank score as the derivative of the accumulated log process at zero (Cochran, 1954; Mantel and Haenszel, 1959). This identity explains how classical weights enter the e-process, but does not equate its rejection rule with the classical test. We also derive a relative-weight criterion for a fixed-margin local score shift and distinguish it from log-growth and boundary-crossing objectives.

The other contribution is an explicit observation rule for combining exact failures and completed-window detections. We retain only subjects observed through the end of a window, unlike the convention that events precede censoring in ter Schure et al. (2024, Section 3.3). A sufficient model for exact comparisons has common null event intensities within each stratum; for completed windows, conditional exchangeability of the interval event indicators gives the required allocation law. Randomization or a chosen stratification alone is insufficient. When these laws hold given the combined history, a study with mixed observation schemes, as in Zhao et al. (2008), can retain both kinds of records in one process without imputing event times. Keeping exact timing and retaining large comparison sets are competing considerations for power.

Section 2 starts with a single-failure calculation and states the observation assumptions. Section 3 separates validity, the score identity and the restricted local weighting criterion. The simulations in Section 4 lead with specifications held fixed across alternatives, then examine how alternative-specific coefficient selection changes the conclusions. Section 5 treats the clinical analyses as exploratory calculations under working observation models. Proofs, full numerical grids and additional diagnostics are in the appendices.

2 Proposed method

We partition subjects into strata for comparison. Each stratum groups subjects with the same prespecified prognostic categories and compatible observation schemes and time scales.

Consider one exact failure in a stratum containing N_{1j} treatment subjects and N_{0j} controls at risk. Under a common null event intensity, each subject is equally likely to be the one who fails. The probability of a treatment failure is therefore $\pi_j = N_{1j}/N_j$, where $N_j = N_{0j} + N_{1j}$. Let Y_j indicate a treatment failure. Before using its group identity, choose a positive conditional-odds alternative θ_j. Its likelihood ratio against the null is

$$e_j = \frac{\theta_j^{Y_j}}{1 - \pi_j + \pi_j \theta_j}, \qquad \theta_j = \exp(\eta w_j). \tag{1}$$

Here w_j is a survival weight and η sets the strength and direction of the alternative. With equal risk-set sizes and $\theta_j = 1/2$, a control failure multiplies the evidence by 4/3, whereas a treatment failure multiplies it by 2/3. Under the null, their average is one. Multiplying such factors accumulates evidence that can increase or decrease after each failure. For a completed interval containing several newly detected events, the same argument uses a hypergeometric distribution instead of a Bernoulli distribution.

2.1 The null hypothesis and comparison tables

The null concerns event allocation within each comparison, conditional on the observed history and recorded prognostic covariates. For exact failures, a sufficient model gives subjects in the current stratum a common event intensity. For grouped assessments, the newly detected event indicators must be conditionally exchangeable within a common completed window. Marginal equality of the two survival distributions alone does not establish these conditions. Event risks may differ between strata, but randomization or a prespecified stratification alone does not remove risk differences within a stratum.

For subject i, let $G_i \in \{0, 1\}$ denote control or treatment, X_i the baseline covariates, and T_i, C_i the event and censoring times. We allow $T_i = \infty$. A fixed map $\mathsf{s}(X_i) \in \{1, \dots, K_X\}$ assigns prognostic strata before follow-up and is applied identically to both groups. Subjects with incompatible observation schemes or time scales are also separated into monitoring strata. The unstratified analysis has $K_X = 1$. Formal probability-space and observation-record conventions are collected in Appendix A.1.

At an exact failure, use the risk set immediately before that failure in its monitoring and prognostic stratum. At an interval assessment, use subjects known event-free at ℓ_j and observed through r_j for a common completed interval $(\ell_j, r_j]$, where $0 \le \ell_j < r_j < \infty$. First detections at r_j are its events. This includes left-censored first detections when $\ell_j = 0$. Exclude intervals not completed at the common endpoint r_j. Inclusion based on terminal event status obtained through another ascertainment mechanism requires a separate justification of Condition 2.1 for the retained comparison. Each retained event enters once, when its occurrence first becomes known.

Index these comparisons chronologically by j. Let $\mathsf{R}_j^{\text{obs}}$ be the eligible set before restricting to prognostic stratum k_j. Write R_j for the restricted set. The event set A_j contains the newly observed events in this stratified set R_j, so that

$$\mathsf{R}_j = \{i \in \mathsf{R}_j^{\text{obs}} : \mathsf{s}(X_i) = k_j\}, \qquad \mathsf{A}_j \subseteq \mathsf{R}_j,$$

and the group sizes and event counts are

$$\begin{aligned} N_{gj} &= \sum_{i \in \mathsf{R}_j} \mathbf{1}\{G_i = g\}, \quad g \in \{0,1\}, \qquad N_j = N_{0j} + N_{1j}, \\ M_j &= |\mathsf{A}_j|, \qquad Y_j = \sum_{i \in \mathsf{A}_j} G_i. \end{aligned} \tag{2}$$

The sets and counts depend on the observed history and fixed stratification. Empty comparison sets are omitted. Every update has the following table:

	Newly observed event	No newly observed event	Total
Treatment	Y_j	$N_{1j} - Y_j$	N_{1j}
Control	$M_j - Y_j$	$N_{0j} - M_j + Y_j$	N_{0j}
Total	M_j	$N_j - M_j$	N_j

An exact failure has $M_j = 1$; a completed assessment window may have several events. A window with no events contributes no evidence, but its negative assessments establish eligibility for later comparisons.

2.2 Information and the conditional null distribution

Weights and alternatives may use past observations, the current comparison set and its total event count, but not the current allocation of those events between groups. Let $\mathfrak{F}_0$ contain group labels, recorded covariates, the fixed stratification and design information. The filtration $\mathfrak{F}_j$ contains all observations incorporated through update j. If O_j records the observation scheme, failure time or interval, and monitoring and prognostic strata, define

$$\mathfrak{K}_j = \mathfrak{F}_{j-1} \vee \sigma(O_j, \mathsf{R}_j), \qquad \mathfrak{G}_j = \mathfrak{K}_j \vee \sigma(M_j). \tag{3}$$

Thus $\mathfrak{G}_j$ adds the margins without adding the event identities. We require

$$\mathfrak{F}_{j-1} \subseteq \mathfrak{K}_j \subseteq \mathfrak{G}_j \subseteq \mathfrak{F}_j, \qquad \mathfrak{G}_j \vee \sigma(\mathsf{A}_j) \subseteq \mathfrak{F}_j.$$

Entry, censoring and assessment information arriving between updates is incorporated before the next comparison. Simultaneous updates in disjoint strata use a prespecified order, excluding unprocessed event allocations until their turn. The conditional allocation null below must hold given earlier observations from every stratum, including allocations already processed at the same time. A prespecified order alone does not establish that null. This specifies the information an analysis may use; it does not require physical release of data in separate stages.

Write $\mathcal{P}_0$ for the family of laws satisfying the following conditional null, and $\mathbb{P}_0, \mathbb{E}_0$ for probability and expectation under any such law. Conditional statements hold almost surely.

Condition 2.1 (Conditional allocation null)**.** *For every* $\mathbb{P}_0 \in \mathcal{P}_0$*, conditional on* $\mathfrak{G}_j$*, each subset of* R_j *containing* M_j *subjects has the same probability of being the event set* A_j*,* $\mathbb{P}_0$*-almost surely.*

Consequently, $Y_j \mid \mathfrak{G}_j \sim \text{Hypergeom}(N_j, N_{1j}, M_j)$, with support

$$\mathcal{Y}_j = \{\max(0, M_j - N_{0j}), \dots, \min(M_j, N_{1j})\}$$

and mass function

$$p_{0j}(y) = \frac{\binom{N_{1j}}{y}\binom{N_{0j}}{M_j - y}}{\binom{N_j}{M_j}}, \qquad y \in \mathcal{Y}_j. \tag{4}$$

This is the conditional comparison underlying the CMH score. In a completed-window model with independent event indicators and common probability p_j, conditioning on M_j cancels $p_j^{M_j}(1-p_j)^{N_j - M_j}$. The unknown event probability may therefore differ between strata and windows without being estimated.

2.3 Observation assumptions

For exact failures, a common null event intensity within the current monitoring and prognostic stratum makes the identity of the failing subject uniform over the current risk set conditional on the continuous-time history immediately before the failure, together with its time, provided censoring and entry preserve the risk-set construction (Andersen et al., 1993; Fleming and Harrington, 1991). This implies Condition 2.1 when $\mathfrak{G}_j$ contains no information beyond that history and failure time; the argument following Proposition A.5 gives the transfer to the update filtration. If an earlier update at the same time has revealed another allocation, the uniform law must also hold conditional on that additional information. The exact-event construction assumes no simultaneous failures within a stratum. Tied failures can instead be one grouped update only if, given $\mathfrak{G}_j$, all event subsets of the observed size are equally likely under the null. An event–censoring tie counts as an observed event; censoring without an event removes the subject from subsequent risk sets.

For a completed interval, let $D_{ij} = \mathbf{1}\{\ell_j < T_i \le r_j\}$ for $i \in \mathsf{R}_j$. Conditional exchangeability of $(D_{ij} : i \in \mathsf{R}_j)$ given $\mathfrak{K}_j$ is sufficient. Conditional independence with a common $\mathfrak{K}_j$-measurable interval event probability is a stronger sufficient condition:

$$\mathbb{P}_0(D_{ij} = 1 \mid \mathfrak{K}_j) = p_j, \qquad i \in \mathsf{R}_j. \tag{5}$$

Here p_j is an interval event probability, not the treatment risk-set fraction π_j in (1). Independent censoring preserves this model under the conditions in Proposition A.6. Merely rounding different visit intervals to the same endpoints does not establish exchangeability.

Stratification must make the common-risk model credible. Coarse strata can retain prognostic heterogeneity; fine strata can leave few informative treatment–control comparisons. The construction does not supply unrestricted adjustment for continuous covariates or unmeasured confounding. Attendance or loss to follow-up related to event risk requires an observation model or sensitivity analysis. Existing inverse-probability adjustments cannot simply be inserted without a new conditional e-value argument (Heitjan and Rubin, 1991; Gill et al., 1997; Robins and Finkelstein, 2000).

2.4 Evidence updates

Choose any $\mathfrak{G}_j$-measurable probability kernel q_j supported on $\mathcal{Y}_j$, with $\sum_{y\in\mathcal{Y}_j} q_j(y) = 1$. Define

$$e_j = \frac{q_j(Y_j)}{p_{0j}(Y_j)}, \qquad E_J = \prod_{j=1}^{J} e_j, \qquad E_0 = 1. \tag{6}$$

Normalization gives $\mathbb{E}_0(e_j \mid \mathfrak{G}_j) = 1$. Iterated expectation then makes E_J a nonnegative null martingale. Its value at a fixed update, or at an almost surely finite stopping time, is an e-value: its null expectation is at most one (Vovk and Wang, 2021; Ramdas and Wang, 2025). Theorem 3.1 gives the corresponding guarantee for repeated monitoring. No assumption that q_j is the true alternative distribution is needed for validity. If the margins determine Y_j, the factor is one; after the final available update, all further factors are also set to one.

We encode a classical survival weight through an exponential tilt. For a finite $\mathfrak{G}_j$-measurable w_j and finite prespecified η, set

$$q^{(w)}_{\eta,j}(y) = \frac{p_{0j}(y)\exp(\eta w_j y)}{\sum\limits_{u\in\mathcal{Y}_j} p_{0j}(u)\exp(\eta w_j u)}, \qquad y \in \mathcal{Y}_j. \tag{7}$$

This is Fisher's noncentral hypergeometric law with conditional-odds parameter $\exp(\eta w_j)$ (Fog, 2008). Its factor and cumulative process are

$$e^{(w)}_j(\eta) = \frac{\exp(\eta w_j Y_j)}{\sum\limits_{u\in\mathcal{Y}_j} p_{0j}(u)\exp(\eta w_j u)}, \tag{8}$$

$$E^{(w)}_J(\eta) = \prod_{j=1}^{J} e^{(w)}_j(\eta). \tag{9}$$

For nonnegative weights, $\eta < 0$ targets fewer treatment events than expected under the conditional null. This coefficient is a working conditional log-odds parameter, not generally a population log hazard ratio. The exponential tilt is a particular admissible numerator within the construction based on conditional likelihood ratios of ter Schure et al. (2024).

2.5 Weights and alternative coefficients

The weight changes the relative emphasis over follow-up. Table 1 gives the choices considered here. For weights based on survival, $\widehat{S}_{j-1}$ is the current stratum's pooled estimate before using the current event allocation. On a common consecutive assessment grid, initialize $\widehat{S}_{k,0} = 1$ and update stratum k_j by

$$\widehat{S}_{k_j,j} = \widehat{S}_{k_j,j-1}\left(1 - \frac{M_j}{N_j}\right), \tag{10}$$

leaving the other strata unchanged. This is the Kaplan–Meier update for exact failures and its grouped analogue for common assessment windows. Comparisons released together use the estimates available before that release; subsequent pooling must be prespecified. Other estimators are admissible if the resulting weights are finite and $\mathfrak{G}_j$-measurable. The lower bound $s_{\min} > 0$ prevents an unbounded inverse weight when estimated survival is small.

Table 1: Classical weights and their intended time emphasis. These choices specify working alternatives; their performance also depends on the coefficient and scale.

Procedure	Weight w_j	Intended emphasis
Logrank/CMH (LR)	1	Approximately proportional event intensities
Gehan–Breslow (GB)	proportional to N_j	Earlier events, when risk sets are larger
Peto–Peto (PP)	$\widehat{S}_{j-1}$	Earlier and middle follow-up
Fleming–Harrington (FH)	$\widehat{S}_{j-1}^{\rho}(1-\widehat{S}_{j-1})^{\gamma}$	Early effects for $\rho > 0$, $\gamma = 0$; delayed effects for $\rho = 0$, $\gamma > 0$
Gray–Tsiatis (GT)	$1/\max(\widehat{S}_{j-1}, s_{\min})$	Greater relative emphasis on later differences

Fleming–Harrington parameters satisfy $\rho, \gamma \geq 0$, with $0^0 = 1$. With the same preceding survival estimate in (10), PP is this family at $(\rho, \gamma) = (1, 0)$; the simulation column FH uses $(0, 1)$. The logrank and Gray–Tsiatis motivations are examined in Section 3, including the difference between right-endpoint and preceding-survival weights for finite intervals.

The coefficient and weight scale must be chosen together: multiplying every weight by $a > 0$ and dividing η by a leaves (7) unchanged. External estimates or independent design simulations can guide their choice. Such simulations should match the planned observation schedule, censoring and prognostic strata. Mean terminal log evidence, crossing probability by the planned end of follow-up and expected monitoring duration are different objectives. To estimate the last quantity, average first-crossing times across all datasets, assigning the full planned duration to every dataset that does not cross. The median among rejecting datasets alone does not measure expected duration.

When the effect is uncertain, a prespecified mixture can combine complete processes. For example, logrank and Gray–Tsiatis processes may be combined as

$$E_J^{\text{mix}} = \lambda E_J^{(1)} + (1-\lambda) E_J^{(2)}, \qquad 0 \leq \lambda \leq 1.$$

A fixed convex combination of null martingales is again a null martingale. The same argument applies to a fixed mixture over coefficients. A finite coefficient η_j learned during follow-up may instead use completed earlier updates, so that it is $\mathfrak{F}_{j-1}$-measurable. Maximizing the already accumulated process over weights or coefficients does not have this justification.

2.6 Practical analysis procedure

Step 1. State the event comparison and the observation and common-risk assumptions supporting Condition 2.1. Choose prognostic and monitoring strata that make those assumptions credible, then specify the direction, weight scale and coefficient or fixed mixture.

Step 2. At each exact failure or completed interval, form the eligible within-stratum comparison and its margins. Use (4) for the null law, with each first event counted once.

Step 3. Evaluate the factor using a weight and alternative that are $\mathfrak{G}_j$-measurable. Multiply factors in the specified time order, incorporating all comparisons released at a monitoring time before inspecting the process. For a mixture, average the complete component processes with the fixed mixing probabilities.

Step 4. Reject on the first monitored crossing of $1/\alpha$, or 20 when $\alpha = 0.05$. Report the evidence path, first crossing, maximum and final value with the comparison and weighting specifications.

Evidence may decline after a valid crossing without reversing the decision. Monitoring regimes may change only according to a prespecified rule or completed earlier observations, preserving the conditional null and avoiding repeated use of an event or overlapping follow-up segment.

3 Validity, score connection and local weight choice

The validity argument is the standard normalization of conditional likelihood ratios. The score identity explains the relation to weighted survival tests. The final result is a restricted design calculation for fixed margins, not an optimization of a survival study's stopping rule. Proofs and observation-specific sufficient conditions are in Appendix A.

Theorem 3.1 (Validity by conditional normalization)**.** *Suppose Condition 2.1 holds at every observation update and* q_j *is a* $\mathfrak{G}_j$*-measurable probability kernel supported on* $\mathcal{Y}_j$*. Then* $(E_J, \mathfrak{F}_J)_{J\geq 0}$*, with* E_J *defined in* (6) *and held constant after the final update, is a nonnegative martingale with* $E_0 = 1$ *under every* $\mathbb{P}_0 \in \mathcal{P}_0$*. Consequently, for every* $\alpha \in (0,1)$,

$$\mathbb{P}_0\left(\sup_{J\geq 0} E_J \geq \alpha^{-1}\right) \leq \alpha. \tag{11}$$

Ville's inequality (Ville, 1939) permits monitoring after every valid update without fixing their number in advance. Confidence sequences for the conditional-odds coefficient require the stronger assumption that its model family contains the true conditional laws; see Appendix A.8.

The hypergeometric conditional mean and variance are

$$\mu_j = M_j \frac{N_{1j}}{N_j}, \qquad v_j = \begin{cases} M_j \dfrac{N_{1j}}{N_j}\left(1 - \dfrac{N_{1j}}{N_j}\right)\dfrac{N_j - M_j}{N_j - 1}, & N_j > 1, \\ 0, & N_j \leq 1. \end{cases}$$

Theorem 3.2 (Score and curvature at the null)**.** *Assume that each* w_j *is finite,* $\mathfrak{G}_j$*-measurable and does not depend on* η*. For the exponentially weighted process in* (9),

$$\frac{\partial}{\partial \eta} \log E_J^{(w)}(\eta)\bigg|_{\eta=0} = \sum_{j=1}^{J} w_j (Y_j - \mu_j), \tag{12}$$

and

$$-\frac{\partial^2}{\partial \eta^2} \log E_J^{(w)}(\eta)\bigg|_{\eta=0} = \sum_{j=1}^{J} w_j^2 v_j. \tag{13}$$

Write U_J and V_J for the right-hand sides of (12) and (13), respectively. For a fixed finite sequence of updates and weights independent of η, the finite-support cumulant functions give the pathwise expansion

$$\log E_J^{(w)}(\eta) = \eta U_J - \frac{\eta^2}{2} V_J + o(\eta^2), \qquad \eta \to 0. \tag{14}$$

The linear term is the weighted CMH or logrank score; the quadratic term penalizes the magnitude of the working alternative. This derivative identity does not make the finite-coefficient e-test equivalent to the classical score test or give it the same rejection region. If the score increments are square-integrable, $\mathrm{Var}_0(U_J) = \mathbb{E}_0(V_J)$. With deterministic counts and weights, this reduces to $\mathrm{Var}_0(U_J) = V_J$.

For local weight selection, fix an exogenously specified finite sequence of deterministic margins $(N_{0j}, N_{1j}, M_j)_{j=1}^{J}$. Let Y_j be adapted to $(\mathfrak{H}_j)_{j=0}^{J}$, with

$$\mathbb{E}_0(Y_j \mid \mathfrak{H}_{j-1}) = \mu_j, \qquad \mathrm{Var}_0(Y_j \mid \mathfrak{H}_{j-1}) = v_j.$$

Under local laws $(\mathbb{P}_\varepsilon)_{\varepsilon \geq 0}$ with null law $\mathbb{P}_0$, suppose deterministic coefficients c_j satisfy

$$\mathbb{E}_\varepsilon(Y_j) = \mu_j + \varepsilon c_j v_j + r_j(\varepsilon), \qquad r_j(\varepsilon) = o(\varepsilon), \tag{15}$$

for each j as $\varepsilon \downarrow 0$.

Theorem 3.3 (Relative weights maximizing a fixed-margin local score shift)**.** *Assume* $\sum_{j=1}^{J} c_j^2 v_j > 0$. *For deterministic finite weights satisfying* $\sum_{j=1}^{J} w_j^2 v_j > 0$, *define* $U_J(w) = \sum_{j=1}^{J} w_j (Y_j - \mu_j)$. *Then*

$$\frac{\mathbb{E}_\varepsilon\{U_J(w)\}}{\mathrm{Var}_0\{U_J(w)\}^{1/2}} = \varepsilon \frac{\sum_{j=1}^{J} w_j c_j v_j}{(\sum_{j=1}^{J} w_j^2 v_j)^{1/2}} + o(\varepsilon). \tag{16}$$

The coefficient of ε *is maximised by* $w_j = \kappa c_j$ *whenever* $v_j > 0$, *for any constant* $\kappa > 0$.

The result concerns relative weights because their common scale cancels. If $c_j \leq 0$, nonnegative weights proportional to $-c_j$ maximize the magnitude of the negative local shift. The calculation does not optimize crossing probability, expected stopping time or estimated weights under randomly evolving risk sets.

Appendix A.7 gives nearly constant c_j under approximately proportional population hazards over short intervals, motivating logrank weights. Its cure-rate calculation uses

$$S_{\mathrm{pop},c}(t) = c + (1-c) S_{\mathrm{sus}}(t), \tag{17}$$

where $c \in [0, 1)$ is the probability of never experiencing the event and S_{sus} is survival among subjects who eventually experience it. Within one stratum, suppose $S_{\mathrm{sus}}(\ell_j) > S_{\mathrm{sus}}(r_j)$ and $S_{\mathrm{pop},c}(r_j) > 0$. Increasing c while fixing susceptible survival gives $c_j = -1/\{(1 - c) S_{\mathrm{pop},c}(r_j)\}$. Thus the fixed-margin relative weight is inverse survival at the right endpoint, not at the start of the interval. On a common consecutive grid, replacing estimates by $S = S_{\mathrm{pop},c}$ and ignoring truncation gives

$$w_j^* = \frac{1}{S(r_j)}, \qquad w_j^L = \frac{1}{S(\ell_j)}, \qquad \frac{w_j^L}{w_j^*} = \frac{S(r_j)}{S(\ell_j)} = 1 - p_j, \tag{18}$$

where $p_j = 1 - S(r_j)/S(\ell_j)$ is the interval event probability. The implemented Gray–Tsiatis rule uses preceding estimated survival, so it approximates this relative weight when interval event probabilities are small. For coarse intervals, the ratios can vary appreciably and Theorem 3.3 does not establish local optimality of that rule. Estimation and truncation introduce further differences but do not affect validity when weights remain finite and $\mathfrak{G}_j$-measurable. If cure probabilities differ between strata, a common additive cure-rate change also retains the stratum factor $1/(1 - c_k)$. Finally, proportional hazards among susceptible subjects need not imply proportional population hazards, and this weighting rationale neither fits a cure model nor identifies cured subjects.

Table 2: Specification regimes used in the simulations. Pilot selection maximizes mean terminal log evidence. All pilots are independent of evaluation data.

Experiment	Weight and normalization	Coefficient choice and quantities held fixed
Fixed specifications	Pooled survival before each update; scale from 1,000 S1 null datasets at $n = 500$, separately by observation scheme	$\eta = \log(0.7)$, or fixed mixtures; the same scales and coefficients at every sample size and alternative
Alternative-specific weights	Pooled survival before each update; scale from 1,000 null pilots for each design	1,000 alternative pilots choose each weight's coefficient for that design
Primary S1/S2 and mixed observations	LR in S1; GT based on the control survival curve supplied by the simulation design in S2	1,000 alternative pilots choose the coefficient for the scenario, sample size and analysis
Allocation, visits and effect sensitivity	LR for timing effects; design-survival GT for cure effects	Reference coefficient chosen at $n = 500$, then retained across the specified design grid; effect-specific selection is a separate comparison

4 Simulation study

The main design question is how a weight performs when its coefficient must be chosen before the treatment effect is known. We therefore begin with specifications held fixed across alternatives, then examine what changes when independent pilot data are generated under the alternative being evaluated. All reported rejection probabilities use 5,000 evaluation datasets, independent of the pilots, at sample sizes $n \in \{100, 200, 500, 1000, 2000\}$. Unless specified otherwise, treatment assignment has probability 1/2, with random group sizes. Exact and interval analyses of a given simulated dataset share its event and censoring times.

The five alternatives distinguish a susceptible event-time difference with equal cure probabilities (S1), a cure-probability difference with equal susceptible hazards (S2), an early effect (S3), a delayed effect (S4), and a combination of timing and cure effects (S5). The event and observation models, including all parameters, appear in Table B.8 and Appendix B.5. Table 2 separates the sources of information used to choose each procedure.

4.1 Weight choice with the specification held fixed

For each observation scheme, 1,000 independent S1 null datasets at $n = 500$ normalize LR, GB, PP, FH and GT to the same null-pilot estimate of terminal score variance. Appendix B.5.5 gives the raw weights, including the GB denominator, and the normalization formula. All five processes use $\eta = \log(0.7)$, with pooled survival evaluated before the current allocation, $(\rho, \gamma) = (0, 1)$ for FH and $s_{\min} = 0.12$ for GT. Cmix averages three LR processes at $\eta \in \{\log(0.5), \log(0.7), \log(0.9)\}$ with equal probabilities. Wmix averages the nine combinations of these coefficients and LR, FH and GT. The pilot normalization multipliers, coefficients and mixing probabilities are retained at every evaluation sample size and alternative; the raw GB denominator is determined within each dataset.

The unit-weight fixed-alternative process agrees with that of ter Schure et al. (2024) when the comparison sets agree, and is reported once as LR. Their prequential procedure, PQ, estimates its coefficient from strictly earlier observations to predict the next allocation. It can learn either higher or lower treatment event odds. Our one-sided adaptation, PQ^{-}, uses the fitted coefficient when negative and zero otherwise. The exact-event implementation follows their procedure; the grouped version extends the fit to our completed-window comparison sets. Appendix B.4 specifies the likelihood, smoothing and initialization for both versions.

Table 3 gives power at $n = 500$. FH retains a substantial advantage for the delayed effect S4 under interval observation: power is 0.6394, compared with 0.2848 for LR. The paired difference is 0.3546 with Monte Carlo standard error 0.0069. Mean capped stopping times are 6.5376 and 7.3891 years, respectively, a paired difference of -0.8515 years with standard error 0.0187. Here the late-emphasis weight both rejects more often and reduces average follow-up. In contrast, for the early effect S3, LR power is 0.7078 and PP power is 0.6966. The early-emphasis weight does not improve the observed power under this fixed specification.

Table 3: Power for fixed practical designs and prequential comparators at total $n = 500$.

Scenario	Obs.	LR	GB	PP	FH	GT	Cmix	Wmix	PQ	PQ^{-}
S1	Interval	0.775	0.690	0.734	0.500	0.765	0.677	0.676	0.466	0.507
S1	Exact	0.821	0.729	0.774	0.725	0.832	0.739	0.767	0.576	0.633
S2	Interval	0.823	0.593	0.689	0.834	0.890	0.725	0.797	0.608	0.647
S2	Exact	0.826	0.600	0.693	0.891	0.906	0.740	0.834	0.540	0.626
S3	Interval	0.708	0.684	0.697	0.275	0.651	0.611	0.590	0.413	0.442
S3	Exact	0.780	0.734	0.751	0.497	0.748	0.688	0.696	0.503	0.566
S4	Interval	0.285	0.077	0.111	0.639	0.528	0.191	0.443	0.137	0.162
S4	Exact	0.294	0.085	0.115	0.653	0.557	0.204	0.456	0.105	0.150
S5	Interval	0.777	0.680	0.706	0.650	0.806	0.675	0.685	0.508	0.547
S5	Exact	0.807	0.701	0.727	0.800	0.860	0.721	0.767	0.523	0.602

The weight mixture recovers part of FH's advantage in S4, whereas mixing LR coefficients does not. In interval-observed S4, Wmix has power 0.4432, between LR and FH, whereas Cmix has power 0.1914. Under interval observation, Wmix has lower estimated power than LR in S1–S3 and S5 at $n = 500$. PQ and PQ^{-} have powers 0.1368 and 0.1616 in S4; fitting the coefficient during follow-up does not recover the FH result in this experiment. The largest estimated null crossing probability among the 450 e-process comparisons is 0.0486. Full power and stopping grids, including Table B.12, are retained in the supplement.

4.2 Weight and coefficient selection answer different questions

The alternative-specific experiment chooses a coefficient separately for each weight by maximizing mean terminal log evidence on 1,000 independent alternative pilots. In interval-observed S3 at $n = 500$, it gives PP power 0.668 and LR power 0.595 (Table B.11). This ordering reverses under the fixed specifications above. Because both normalization and coefficient selection differ between experiments, the reversal cannot be attributed to the time emphasis of PP alone. The observed early-effect advantage concerns the joint choice of weight and coefficient with knowledge of the alternative. The FH advantage in S4 is less dependent on this distinction: it appears under both alternative-specific selection and the fixed specifications.

Independence of pilot and evaluation data prevents reuse of evaluation outcomes for selection, but it does not remove the information supplied by choosing a design under the true alternative. This is particularly relevant to the primary S2 and mixed-observation experiments, whose GT weights use the control survival curve fixed by the simulation design. Those experiments assess the specified design-informed alternative, not estimation of that survival curve from trial data.

The local score criterion, mean terminal log evidence and boundary-crossing probability are different objectives. Table B.2 makes the distinction concrete. For $n = 500$, interval observation and susceptible hazard ratio 0.80 in the latency family, replacing the reference coefficient by the effect-specific log-growth choice increases mean terminal log evidence from 0.742 to 1.238, while power falls from 0.287 to 0.193. A less aggressive coefficient can improve the average final log evidence without producing as many paths that reach 20 before follow-up ends. Selecting for log growth therefore does not select for rejection within the available duration. For study planning, the intended effect range should be evaluated using crossing probability and mean capped duration as well as the objective used to choose the coefficient.

4.3 Power and duration under the specified monitoring rules

For the alternative-specific primary comparison, E rejects when the running e-process first reaches 20. Fixed applies the corresponding weighted score test once at the planned endpoint. OBF uses the boundary $-1.96/\sqrt{t_k}$ at the planned calendar looks, with information fractions t_k estimated from independent null pilots. Repeated applies the terminal score critical value after every update without a monitoring adjustment. Full definitions, pilot settings and the common-grid generalized-logrank benchmark appear in Appendix B.5.

The power difference is large at small sample sizes. With interval observations in S1 at $n = 100$, power is 0.077 for the e-process and 0.376 for Fixed. Figure 1 shows that the difference narrows as the sample size increases under these alternatives. At $n = 500$ in the same setting, the respective powers are 0.795 and 0.914. These differences reflect the specified evidence process, log-growth coefficient selection and monitoring rules, not an unavoidable power loss for every anytime-valid procedure. If a single planned analysis is feasible, the optional-monitoring guarantee alone is not a reason to prefer the e-process. When continued monitoring or optional continuation is required, the sample size and working alternative must be assessed against the desired probability of reaching a conclusion.

The OBF null rejection estimates are mostly 0.033–0.042. Its boundary was not calibrated to spend 0.05 over the finite Gaussian information schedule, so comparisons here concern this conservative boundary, not an optimized group-sequential design. In interval-observed S1 at $n = 500$, the e-process has median stopping time 2.00 years among rejections, compared with 3.00 for OBF, but its mean capped time is slightly longer, 3.81 versus 3.74 years. Nonrejecting datasets contribute the full planned duration to this mean. Earlier rejection in selected datasets need not reduce expected follow-up. Fixed has known analysis time 8 years in S1 and 10 years in S2 (Table B.9).

The null curves also distinguish monitoring schemes. Exact observation provides more event-time updates as the sample size increases; grouped observation retains the fixed annual assessment grid and increases the counts within each update. The grouped null crossing estimates remain well below 0.05 and are not monotone in sample size, whereas the

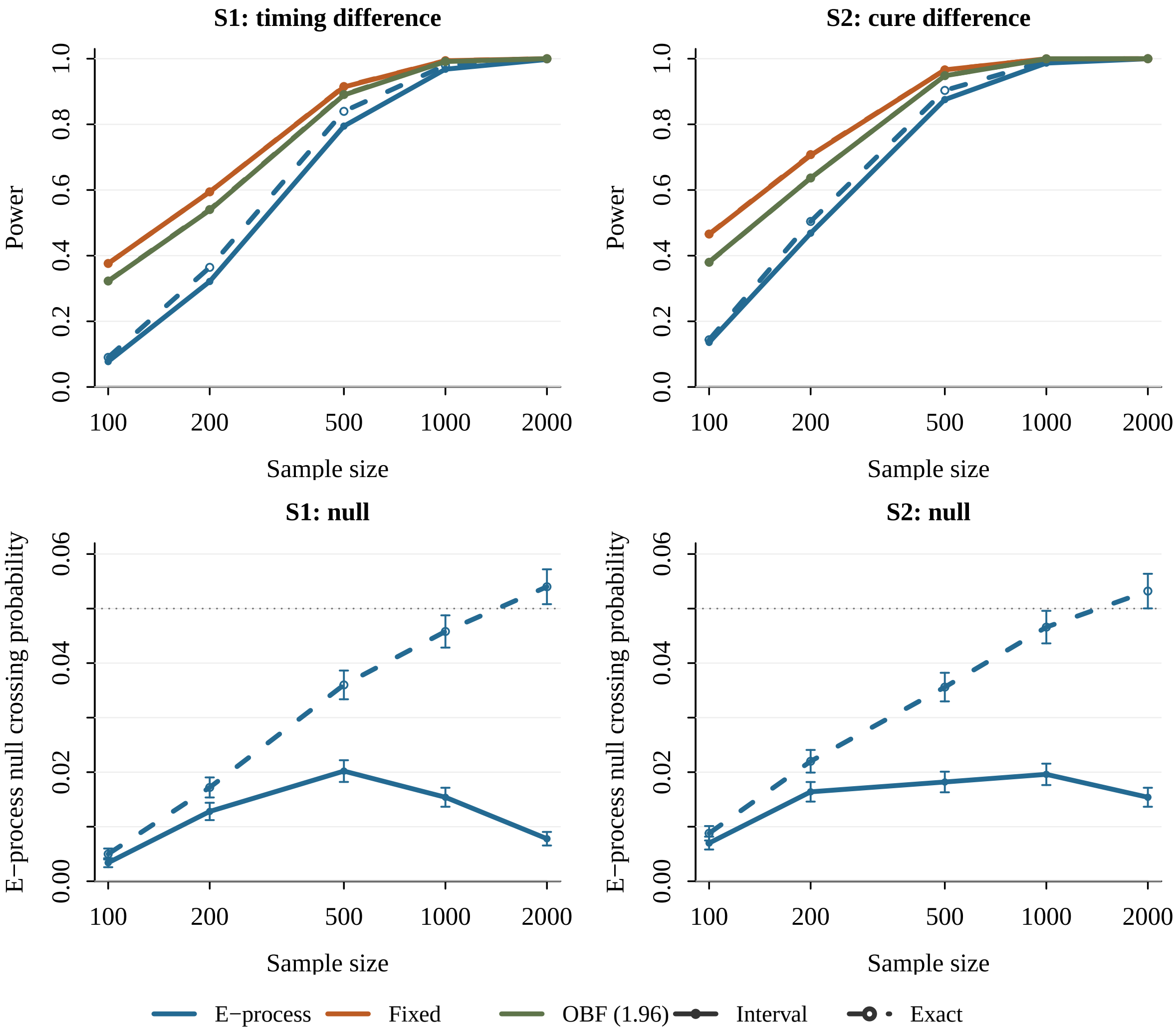

Figure 1: Primary S1 and S2 results by sample size, using alternative-specific log-growth coefficients. Top: power for the e-process, Fixed and the OBF comparator with constant 1.96. Bottom: e-process null crossing probabilities; bars show one Monte Carlo standard error, and the dotted line is 0.05. Solid lines and filled points denote Interval observation; dashed lines and open points denote Exact observation. The sample-size axis is logarithmic. Each estimate uses 5,000 datasets.

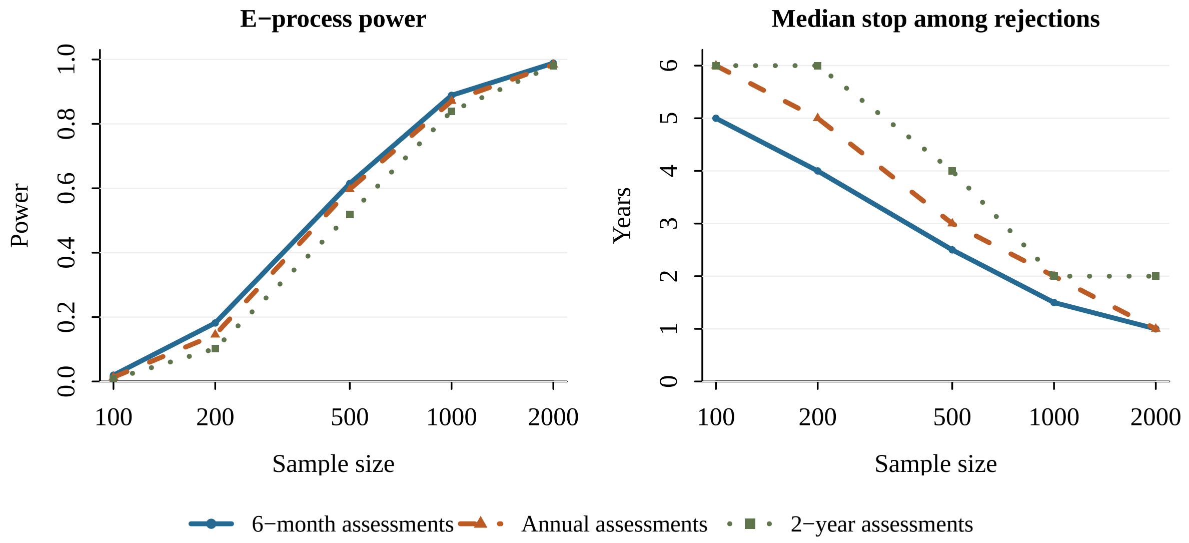

Figure 2: Effect of assessment spacing with a fixed 8-year duration and the same annual-design coefficient. Left: e-process power. Right: median stopping time conditional on rejection, which is not expected study duration. Lines represent 6-month, annual and 2-year assessments. The sample-size axis is logarithmic, and each estimate uses 5,000 alternative datasets.

exact estimates approach 0.05 in these designs. Coefficients are also selected separately by sample size. The plots therefore reflect changes in update granularity and specification, not a failure of the conditional martingale argument or a theorem about monotone convergence to the nominal level. Across primary designs the estimated e-process null rejection ranges from 0.003 to 0.054, compared with 0.125–0.509 for Repeated; Monte Carlo variation must be considered when interpreting estimates near 0.05.

4.4 Assessment schedules and comparison-set quality

The assessment-frequency experiment holds follow-up at 8 years and the event-time model fixed while varying visit spacing over 0.5, 1 and 2 years. It uses susceptible control hazard 0.21, treatment hazard ratio 0.70, cure probability 0.15 in both groups and independent exponential censoring at rate 0.085. The annual-design coefficient selected at $n = 500$ is retained at all sample sizes and spacings. Only completed intervals contribute. Thus changing visit spacing changes both when evidence can arrive and how much follow-up remains unassessed before censoring.

At $n = 500$, e-process power is 0.615, 0.597 and 0.519 at the three spacings, with conditional median stopping times 2.50, 3.00 and 4.00 years (Figure 2). The differences are much smaller near the upper end of the sample-size grid, where power is already high. These results concern the retained annual-design coefficient, not a separate optimum for each schedule. The full grid and the effect-size sensitivity designs are given in Tables B.15 and B.16.

The allocation and censoring grid keeps the coefficient fixed while changing sample size, treatment assignment probability and loss to follow-up (Tables B.13 and B.14). Its e-process null rejection estimates range from 0.000 to 0.052, compared with 0.115–0.447 for Repeated. A different issue arises in the covariate experiment: treatment assignment favours the lower-risk stratum, and subjects enter at different event-free times. Stratifying the completed-interval comparisons gives null rejection between 0.001 and 0.021; pooling gives 0.179–0.842 (Table B.17). The contrast shows why a valid monitoring rule cannot repair a comparison whose conditional null is wrong. The detailed entry-conditioned cure model and all design parameters are retained in the supplement.

4.5 Retaining exact times in a partly interval-censored sample

In S1 and S2, each subject is independently assigned to exact observation with probability $r_{\text{exact}} \in \{0, 0.25, 0.5, 0.75, 1\}$, independently of treatment, event time and censoring. The remaining subjects are assessed annually. The unified analysis combines exact risk-set updates and completed-interval updates, formed within their respective observation subcohorts, in time order. It is compared with using only the exact subcohort, coarsening everyone to annual intervals, and an all-exact benchmark that has additional event-time information. All analyses share the latent datasets. Independent alternative pilots choose log-growth coefficients separately for each analysis and fraction; the all-interval and all-exact coefficients are shared across fractions.

At $n = 500$ and $r_{\text{exact}} = 0.5$, unified power is 0.8190 in S1 and 0.8984 in S2, compared with 0.4924 and 0.6366 when the interval-observed subcohort is discarded. Coarsening the full sample gives 0.7856 and 0.8776. The latter comparison asks a more specific question: whether preserving exact timing compensates for making comparisons within smaller observation subcohorts. At $n = 100$ in S1, the unified estimate at the same fraction is 0.0744, below 0.0792 for all-interval observation. These are descriptive differences, not established paired effects without their paired uncertainty. Exact timing can add information, but smaller risk sets and grouped comparison sets can offset that benefit. Tables B.3 and B.4 retain the complete null and power results; they support examining this tradeoff in the intended study design rather than assuming a fixed ordering.

5 Exploratory clinical applications

5.1 Observation models and analysis specification

The cardiac allograft vasculopathy (CAV) and psoriatic arthritis (PsA) data from `msm` illustrate evidence calculations after irregular visits have been grouped into working assessment intervals. Follow-up is aligned within patients, rather than monitored prospectively across staggered recruitment. The observation assumptions required for exact validity are not established for either cohort. We therefore use 20 as a reference threshold and interpret the weight and grid comparisons descriptively.

Within each application, every weight uses the same comparison tables, prognostic strata and coefficient mixture. LR is the reference weight; GB, PP, FH, GT and the equal LR–GT mixture assess other working alternatives. GB divides the current table size by the eligible baseline count in its prognostic band. PP, FH with $(\rho, \gamma) = (0, 1)$, and GT with $s_{\min} = 0.12$ use the band's survival estimate before the current batch. After all tables at time t, this estimate is multiplied by $1 - \sum_j M_j / \sum_j N_j$, summing over that band's tables at t. It is an estimator for the constructed comparisons, not a Kaplan–Meier estimate from the original visits. Each coefficient mixture averages complete processes. Changing w_j at fixed η changes the magnitude ηw_j as well as its temporal emphasis.

Table 4 distinguishes retained events from informative comparisons. We call a table informative when its hypergeometric conditional variance is positive, requiring both groups and both event and nonevent observations. A table containing only events contributes no comparative evidence even when both groups are present. Full evidence summaries and observed ranges of ηw_j appear in Appendix C.

Table 4: Comparative information after rounding or binning visits. The comparison and event columns show informative/total counts, where informative means positive conditional variance. All 214 CAV detections and 72 PsA progressions are retained at every width, but fewer contribute to a group comparison.

Application	Width q	Retained intervals	Monitoring times	Comparisons	Events
CAV	0.10 years	1477	95	67/521	130/214
CAV	0.25 years	1475	51	70/329	166/214
CAV	0.50 years	1475	31	63/257	175/214
PsA	3 months	202	7	5/16	51/72
PsA	6 months	193	4	4/8	61/72
PsA	12 months	186	2	3/4	71/72

5.2 Cardiac allograft vasculopathy

The `cav` data contain 2,846 records from 622 heart-transplant recipients undergoing approximately yearly angiography (Jackson, 2011; Sharples et al., 2003). We compare 313 recipients with pretransplant ischaemic heart disease, IHD, group 1, and 270 with idiopathic dilated cardiomyopathy, IDC, group 0. The event is first detected mild, moderate or severe CAV, with 128 IHD and 86 IDC detections. We stratify by sex, baseline recipient age below 45, 45–54 or at least 55 years, and donor age below 30 or at least 30 years. The comparison is prognostic and does not remove residual confounding.

Each event interval runs from the last negative to the first positive angiogram. Death before detected CAV censors follow-up at the last negative angiogram, requiring an assumption of noninformative death that is doubtful in this setting. For width q, each endpoint t is rounded to $q\,\mathrm{round}(t/q)$ using R's ties-to-even rule. Intervals with coincident rounded endpoints are omitted; none contains a first CAV detection. A comparison requires agreement of both rounded endpoints and the prognostic categories. Tables ending at rounded time r are combined and inspected at $r + q/2$, after all actual visits assigned to that batch. Exact validity would require a common conditional event probability despite differences between the actual intervals grouped together.

The reference specification uses $q = 0.25$ years and equal weights on $\eta \in \{-\log 2, -\log 1.5, \log 1.5, \log 2\}$. LR first reaches 20 at 4.38 years, peaks at 369.11 at 5.38 years and ends at 34.69. With the same tables, GT ends at 5.14×10^{-15}. The observed GT weights attain the cap 8.33 in informative comparisons, giving ηw_j between -5.776 and 5.776, or conditional odds from approximately 0.0031 to 322.54. LR uses odds from 0.5 to 2. The large difference in terminal evidence therefore compares substantially different alternative magnitudes, as well as different time emphases. Figure 3 also shows declines after earlier threshold crossings. Wider rounding retains more events in informative tables, but changes their composition and the evidence path.

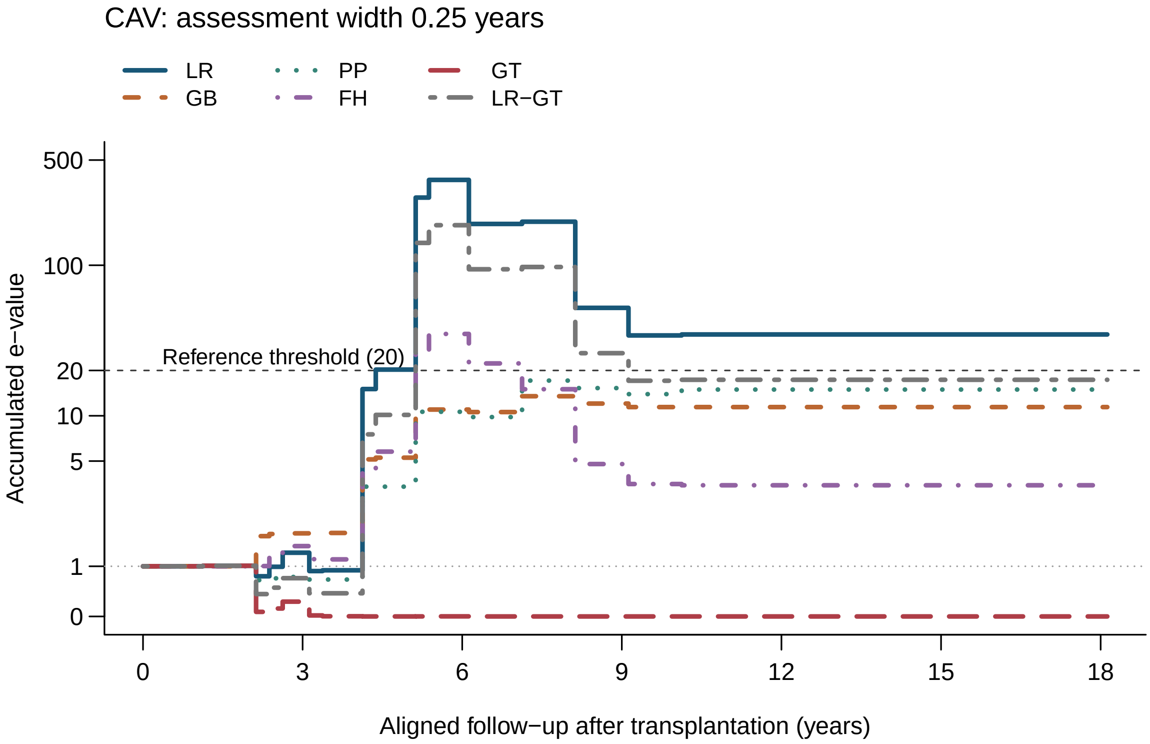

Figure 3: Retrospective CAV evidence at $q = 0.25$ years under the working common-probability assumption for rounded intervals and noninformative censoring at the last negative angiogram. These assumptions are not established for this cohort. Steps occur at fixed upper bin edges on aligned post-transplant follow-up. The line at 20 is a reference threshold. The vertical scale is linear from 0 to 1 and logarithmic above 1.

5.3 Psoriatic-arthritis progression

The `psor` data contain 806 clinic observations from 305 patients (Jackson, 2011; Gladman and Farewell, 1999). Excluding 93 patients with joint damage at their first visit and 27 with missing baseline erythrocyte sedimentation rate (ESR) leaves 185 patients. Of these, 120 have ESR ≥ 15 mm/h, group 0, and 65 have ESR < 15 mm/h, group 1. First progression to at least one damaged joint occurs in 54 and 18 patients, respectively; 66 and 47 are censored at their final damage-free visit. The comparison adjusts for baseline high effusion, present in only 20 eligible patients, including three with ESR < 15 mm/h.

Follow-up begins at the first recorded damage-free visit. In each right-closed nominal window $((k-1)q, kq]$, the last visit is selected by time alone. Follow-up stops at the first retained positive state. Because damage states never decrease, this rule preserves every first progression. Tables group the previous and current nominal bins and high-effusion status, and updates are combined at kq. Actual visit times vary within the bins; attendance, loss to follow-up and treatment may depend on disease activity. Common conditional event probabilities are consequently a working assumption. The comparison is observational, with selection at cohort entry and possible residual confounding.

The reference width is six months, within the source study's 6–12-month assessment protocol. The equal mixture uses $\eta = \log(0.50), \log(0.67), \log(0.80)$. The direction follows a published outcome analysis of the same cohort and is outside the optional-monitoring guarantee. Only four of the eight reference tables are informative, as shown in Table 5. In particular, all seven patients in the $(6, 12]$-month table progress, so these events contribute a factor of one. LR equals 3.19, 8.25, 11.22 and 11.22 at months 6, 12, 18 and 24. GT equals 23.63 by month 18. Its informative-table weights reach 2.326, with $\eta w_j \in [-1.613, -0.223]$, compared with $[-0.693, -0.223]$ for LR. Thus GT's larger contributions at months 12 and 18 use alternatives with lower group-1 event odds as well as greater late emphasis. Because the weights were compared retrospectively, the GT crossing is descriptive. No weight reaches 20 with three- or twelve-month bins.

Table 5: The eight six-month PsA comparison tables. Group columns report events/subjects, with v_j the conditional hypergeometric variance. The last columns decompose the LR and GT mixtures into successive evidence multipliers in the saved table order. Only products over all tables ending at one time are monitored; intermediate multipliers depend on the chosen within-batch ordering.

Window (months)	High effusion	ESR ≥ 15 mm/h	ESR < 15 mm/h	v_j	LR multiplier	GT multiplier
$(0, 6]$	No	32/70	14/47	6.767	3.436	3.436
$(0, 6]$	Yes	5/15	1/3	0.588	0.928	0.928
$(0, 12]$	No	7/28	0/13	1.288	2.588	4.137
$(0, 12]$	Yes	2/2	0/0	0	1	1
$(6, 12]$	No	4/4	3/3	0	1	1
$(0, 18]$	No	2/4	0/2	0.356	1.360	1.792
$(6, 18]$	No	1/1	0/0	0	1	1
$(0, 24]$	No	1/1	0/0	0	1	1

5.4 Stress tests of irregular visits

Fay and Shih (2012) study weighted logrank tests for interval-censored data when assessment distributions depend on treatment, using asymptotic arguments and simulation to examine calibration at a fixed analysis. That setting differs from the repeated-monitoring guarantee under Condition 2.1.

We applied the actual CAV rounding and PsA visit-binning functions to synthetic visits under equal latent event hazards in the two groups. Each of three visit mechanisms has 2,000 replications. For PsA, making group-1 visits earlier within each nominal window changes the observed LR and GT threshold-crossing frequencies from 0.0085 and 0.0090 on the common grid to 0.0540 and 0.0570. Their 95% Wilson intervals, $[0.0449, 0.0648]$ and $[0.0477, 0.0680]$, include 0.05. This shows sensitivity to differential visit timing, without establishing above-nominal crossing probabilities. Group-independent jitter gives 0.0070 and 0.0085, and all CAV estimates remain below 0.05. Appendix C.3 specifies the mechanisms and interval-probability diagnostics. These stress tests do not validate either cohort's observation assumptions.

6 Discussion

Weight choice cannot be separated from the choice of η. In the simulations, Fleming–Harrington weighting retained an advantage for delayed effects when the specification was fixed across scenarios, whereas the apparent early-effect advantage of Peto–Peto weighting depended on selecting a coefficient for that alternative. Increasing mean terminal log

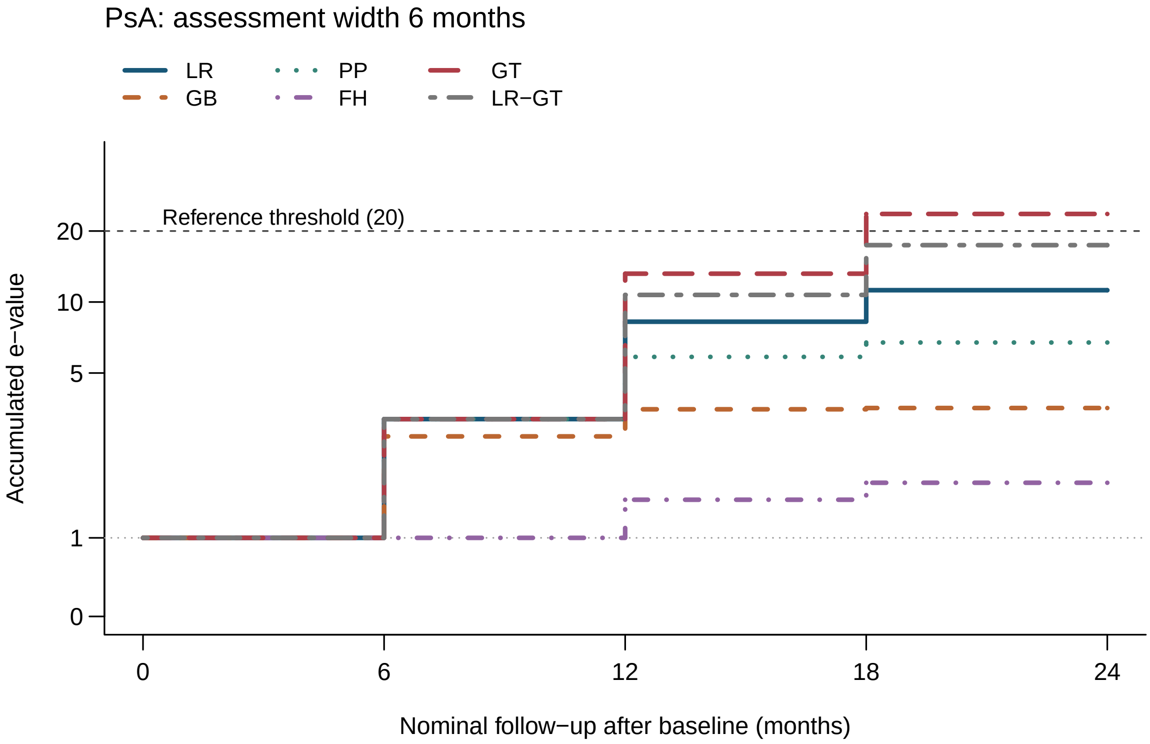

Figure 4: Retrospective PsA evidence at $q = 6$ months under a working common-probability model for nominal visit bins. That model and noninformative attendance are not established for this cohort; the direction was informed by an earlier analysis of the same outcomes. Evidence is reconstructed on aligned follow-up and inspected at the four nominal assessment times. The line at 20 is a reference threshold. The vertical scale is linear from 0 to 1 and logarithmic above 1.

evidence could also reduce the probability of rejection before follow-up ended. A useful design comparison must therefore specify both the anticipated effects and the performance objective. The fixed-margin local score calculation explains relative weighting near the null; it does not select the best finite-duration e-test.

Mixture performance also depends on sample size and the chosen objective. At $n = 100$, Cmix has higher estimated power than LR in all ten scenario–observation combinations in Table B.6. At $n = 2000$ under interval-observed S4, Wmix has higher estimated power than FH, 0.982 versus 0.953, but longer mean capped follow-up, 5.34 versus 5.10 years (Table B.7). These descriptive comparisons do not give a single preferred mixture across study sizes and performance criteria.

The small-sample power losses relative to the terminal score test were substantial for the specified procedures. The e-process buys flexibility to stop or continue without a new type-I error calculation, but that flexibility need not be valuable in a study with one feasible analysis and a firm end date. When monitoring is important, design simulations should report rejection probability and mean capped follow-up across plausible alternatives. Stopping medians conditional on rejection describe the successful replications, not the expected duration of the study. Comparisons with the OBF-type procedure in this paper concern its stated conservative boundary, not an optimized group-sequential design.

The clinical analyses expose a separate difficulty. Rounding or binning irregular visits changes which subjects can be compared and can place subjects with different interval event probabilities in the same table. Retaining each first detection does not retain all comparative information: tables with zero conditional variance contribute a factor of one. The clinical weight comparisons also change the magnitude of the conditional-odds alternative, as well as its timing. Their threshold crossings are exploratory descriptions under working observation assumptions, rather than demonstrations of the exact monitoring guarantee for these cohorts.

The method applies when the conditional allocation model is defensible. Residual prognostic heterogeneity within strata, informative attendance and outcome-related censoring can invalidate it. A rejection with a late-emphasis weight does not identify a cured subgroup without further follow-up and assumptions about susceptible subjects' long-term survival (Maller and Zhou, 1992). Prospective use with staggered recruitment must also respect the order in which observations become available; delayed entry on a common risk-time scale is different from aligning each subject's time since entry (ter Schure et al., 2024, Section 8.2). Extending the construction to history-dependent visits or continuous-covariate adjustment requires a justified conditional allocation law, rather than only a different weight. These observation-model questions are the principal limitation of applying the procedure beyond common completed windows.

Data accessibility statement

Data and code supporting the analyses are available in the accompanying GitHub repository.

AI use statement

OpenAI ChatGPT and Codex were used for exposition, mathematical formulation, proof checking, computational implementation, proofreading, and formatting. The authors take responsibility for all parts of this manuscript.

References

Per Kragh Andersen, Ornulf Borgan, Richard D. Gill, and Niels Keiding. *Statistical Models Based on Counting Processes.* Springer, New York, 1993. (cited on pages 2, 5, and 25)

William G. Cochran. Some methods for strengthening the common chi-squared tests. *Biometrics*, 10(4):417–451, 1954. (cited on page 2)

David R. Cox. Regression models and life-tables. *Journal of the Royal Statistical Society: Series B*, 34(2):187–220, 1972. (cited on page 1)

Michael P. Fay and Pamela A. Shaw. Exact and asymptotic weighted logrank tests for interval censored data: The interval R package. *Journal of Statistical Software*, 36(2): 1–34, 2010. (cited on pages 2, 43, and 45)

Michael P. Fay and Joanna H. Shih. Weighted logrank tests for interval censored data when assessment times depend on treatment. *Statistics in Medicine*, 31(28):3760–3772, 2012. doi:10.1002/sim.5447. (cited on page 18)

Dianne M. Finkelstein. A proportional hazards model for interval-censored failure time data. *Biometrics*, 42(4):845–854, 1986. (cited on page 2)

Thomas R. Fleming and David P. Harrington. *Counting Processes and Survival Analysis*. Wiley, New York, 1991. (cited on pages 5 and 25)

Agner Fog. Sampling methods for Wallenius' and Fisher's noncentral hypergeometric distributions. *Communications in Statistics—Simulation and Computation*, 37(2): 241–257, 2008. (cited on page 6)

Edmund A. Gehan. A generalized Wilcoxon test for comparing arbitrarily singly-censored samples. *Biometrika*, 52(1–2):203–224, 1965. (cited on page 1)

Richard D. Gill, Mark J. van der Laan, and James M. Robins. Coarsening at random: Characterizations, conjectures, counter-examples. *State of the Art in Survival Analysis*, pages 255–294, 1997. (cited on page 5)

Danielle D. Gladman and Vernon T. Farewell. Progression in psoriatic arthritis: role of time-varying clinical indicators. *Journal of Rheumatology*, 26(11):2409–2413, 1999. (cited on page 17)

Robert J. Gray and Anastasios A. Tsiatis. A linear rank test for use when the main interest is in differences in cure rates. *Biometrics*, 45(3):899–904, 1989. (cited on page 1)

David P. Harrington and Thomas R. Fleming. A class of rank test procedures for censored survival data. *Biometrika*, 69(3):553–566, 1982. (cited on page 1)

Daniel F. Heitjan and Donald B. Rubin. Ignorability and coarse data. *The Annals of Statistics*, 19(4):2244–2253, 1991. (cited on page 5)

Christopher H. Jackson. Multi-state models for panel data: The `msm` package for `R`. *Journal of Statistical Software*, 38(8):1–29, 2011. doi:10.18637/jss.v038.i08. (cited on pages 16 and 17)

K. K. Gordon Lan and David L. DeMets. Discrete sequential boundaries for clinical trials. *Biometrika*, 70(3):659–663, 1983. (cited on page 2)

C. G. Law and Ron Brookmeyer. Effects of midpoint imputation on the analysis of doubly censored data. *Statistics in Medicine*, 11(12):1569–1578, 1992. (cited on page 2)

Ross A. Maller and S. Zhou. Estimating the proportion of immunes in a censored sample. *Biometrika*, 79(4):731–739, 1992. (cited on page 20)

Nathan Mantel. Evaluation of survival data and two new rank order statistics arising in its consideration. *Cancer Chemotherapy Reports*, 50(3):163–170, 1966. (cited on page 1)

Nathan Mantel and William Haenszel. Statistical aspects of the analysis of data from retrospective studies of disease. *Journal of the National Cancer Institute*, 22(4):719–748, 1959. (cited on page 2)

Peter C. O'Brien and Thomas R. Fleming. A multiple testing procedure for clinical trials. *Biometrics*, 35(3):549–556, 1979. (cited on page 2)

Aaditya Ramdas and Ruodu Wang. Hypothesis testing with e-values. *Foundations and Trends in Statistics*, 1(1–2):1–390, 2025. (cited on page 6)

James M. Robins and Dianne M. Finkelstein. Correcting for noncompliance and dependent censoring in an AIDS clinical trial with inverse probability of censoring weighted log-rank tests. *Biometrics*, 56(3):779–788, 2000. (cited on page 5)

Linda D. Sharples, Christopher H. Jackson, Jayan Parameshwar, John Wallwork, and Stephen R. Large. Diagnostic accuracy of coronary angiography and risk factors for post-heart-transplant cardiac allograft vasculopathy. *Transplantation*, 76(4):679–682, 2003. doi:10.1097/01.TP.0000071200.37399.1D. (cited on pages 2 and 16)

Eric V. Slud. Sequential linear rank tests for two-sample censored survival data. *The Annals of Statistics*, 12(2):551–571, 1984. (cited on page 2)

Jianguo Sun. A non-parametric test for interval-censored failure time data with application to AIDS studies. *Statistics in Medicine*, 15(13):1387–1395, 1996. (cited on pages 2, 43, and 45)

Jianguo Sun. *The Statistical Analysis of Interval-Censored Failure Time Data*. Springer, New York, 2006. (cited on page 2)

Judith ter Schure, Muriel F. Pérez-Ortiz, Alexander Ly, and Peter D. Grünwald. The anytime-valid logrank test: Error control under continuous monitoring with unlimited horizon. *The New England Journal of Statistics in Data Science*, 2:190–214, 2024. (cited on pages 1, 2, 6, 11, 20, 25, 29, and 38)

Bruce W. Turnbull. The empirical distribution function with arbitrarily grouped, censored and truncated data. *Journal of the Royal Statistical Society: Series B*, 38(3):290–295, 1976. (cited on page 2)

Rosanne J. Turner and Peter D. Grünwald. Safe sequential testing and effect estimation in stratified count data. In *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pages 4880–4893, 2023. (cited on page 2)

Jean Ville. *Étude critique de la notion de collectif*. Gauthier-Villars, Paris, 1939. (cited on page 8)

Vladimir Vovk and Ruodu Wang. E-values: Calibration, combination and applications. *The Annals of Statistics*, 49(3):1736–1754, 2021. (cited on page 6)

Xingqiu Zhao, Qiang Zhao, Jianguo Sun, and Jong S. Kim. Generalized log-rank tests for partly interval-censored failure time data. *Biometrical Journal*, 50(3):375–385, 2008. (cited on page 2)

A Proofs and additional theoretical results

The notation follows Section 2.

A.1 Probability-space and observation-record conventions

All random objects are measurable on a common space $(\Omega, \mathfrak{F})$ under a family of probability measures $\mathcal{P}$, with $\mathcal{P}_0 \subseteq \mathcal{P}$ the conditional null family in Section 2. Subject indices lie in $\mathsf{I}_n = \{1, \ldots, n\}$, group labels in $\{0, 1\}$, baseline covariates in a Borel subset $\mathcal{X} \subseteq \mathbb{R}^p$, and event and right-censoring times in $(0, \infty]$. The fixed stratification map $\mathsf{s} : \mathcal{X} \to \{1, \ldots, K_X\}$ is measurable. These variables, entry times, examination times and comparison sets are measurable, and the discrete observation filtration is contained in $\mathfrak{F}$.

An interval record $(L_i, R_i]$ satisfies $0 \leq L_i < R_i \leq \infty$ and $L_i < T_i \leq R_i$. It includes left censoring when $L_i = 0$, $R_i < \infty$, and right censoring when $R_i = \infty$. An exact record gives T_i itself. Under exact observation with right censoring, write

$$U_i = \min(T_i, C_i), \qquad \Delta_i = \mathbf{1}\{T_i < \infty,\ T_i \leq C_i\}.$$

Thus $\Delta_i = 1$ records an event, including an event–censoring tie; finite U_i with $\Delta_i = 0$ records $(U_i, \infty]$. A random index $\tau \in \mathbb{N}_0 \cup \{\infty\}$ is a stopping time when $\{\tau \leq j\} \in \mathfrak{F}_j$ for every j. Quantities determined by completed earlier updates are $\mathfrak{F}_{j-1}$-measurable; exact-event risk-set weights evaluated before the jump are predictable for the continuous-time filtration used below.

A.2 Proof of anytime validity

Proof of Theorem 3.1. Under Condition 2.1, there are $\binom{N_j}{M_j}$ equiprobable event subsets. Exactly $\binom{N_{1j}}{y}\binom{N_{0j}}{M_j-y}$ contain y treatment subjects, which gives (4). Therefore, conditionally on $\mathfrak{G}_j$, Y_j has mass function p_{0j}, and

$$\mathbb{E}_0\left(e_j \mid \mathfrak{G}_j\right) = \sum_{y \in \mathcal{Y}_j} p_{0j}(y) \frac{q_j(y)}{p_{0j}(y)} = \sum_{y \in \mathcal{Y}_j} q_j(y) = 1.$$

By iterated conditional expectation,

$$\mathbb{E}_0\left(e_j \mid \mathfrak{F}_{j-1}\right) = 1.$$

Because E_{j-1} is $\mathfrak{F}_{j-1}$-measurable,

$$\mathbb{E}_0\left(E_j \mid \mathfrak{F}_{j-1}\right) = E_{j-1} \mathbb{E}_0\left(e_j \mid \mathfrak{F}_{j-1}\right) = E_{j-1}.$$

The process is nonnegative and starts at one. Taking expectations inductively gives $\mathbb{E}_0(E_j) = 1$, so every finite-time value is integrable. Ville's maximal inequality gives (11). □

Corollary A.1 (E-value at a stopping time)**.** *Let $\tau : \Omega \to \mathbb{N}_0 \cup \{\infty\}$ be an almost surely finite stopping time for $(\mathfrak{F}_J)$, meaning that $\{\omega \in \Omega : \tau(\omega) \leq J\} \in \mathfrak{F}_J$ for every J. Then $\mathbb{E}_0(E_\tau) \leq 1$ for every $\mathbb{P}_0 \in \mathcal{P}_0$.*

Proof. For each m, optional sampling at the bounded time $\tau \wedge m$ gives $\mathbb{E}_0(E_{\tau \wedge m}) = 1$. Since $\tau < \infty$ almost surely, $E_{\tau \wedge m} \to E_\tau$ almost surely. Fatou's lemma yields

$$\mathbb{E}_0(E_\tau) \leq \liminf_{m \to \infty} \mathbb{E}_0(E_{\tau \wedge m}) = 1.$$

□

A.3 Proof of the score identity and local expansion

For the factor in (8), write

$$\psi_j(u) = \log \mathbb{E}_0 \{\exp(uY_j) \mid \mathfrak{G}_j\}$$

so that

$$\log e_j^{(w)}(\eta) = \eta w_j Y_j - \psi_j(\eta w_j).$$

Proof of Theorem 3.2. Differentiation gives

$$\frac{\partial}{\partial \eta} \log e_j^{(w)}(\eta) = w_j \left\{ Y_j - \psi_j'(\eta w_j) \right\}.$$

At zero, $\psi_j'(0) = \mathbb{E}_0(Y_j \mid \mathfrak{G}_j) = \mu_j$, the hypergeometric mean. A second differentiation gives

$$-\frac{\partial^2}{\partial \eta^2} \log e_j^{(w)}(\eta) = w_j^2 \psi_j''(\eta w_j).$$

At zero, $\psi_j''(0) = \mathrm{Var}_0(Y_j \mid \mathfrak{G}_j) = v_j$. Summing over j proves both identities.

For the variance interpretation in the main text, let $D_j = w_j(Y_j - \mu_j)$. Then $\mathbb{E}_0(D_j \mid \mathfrak{G}_j) = 0$ and $\mathbb{E}_0(D_j^2 \mid \mathfrak{G}_j) = w_j^2 v_j$. With the refined filtration $\mathfrak{J}_{2j-2} = \mathfrak{F}_{j-1}$, $\mathfrak{J}_{2j-1} = \mathfrak{G}_j$ and $\mathfrak{J}_{2j} = \mathfrak{F}_j$, the cumulative score held constant at odd indices is a martingale. Its predictable quadratic variation at index $2J$ is therefore $\sum_{j=1}^{J} w_j^2 v_j$. Moreover, for $j < k$,

$$\mathbb{E}_0(D_j D_k) = \mathbb{E}_0 \{D_j \mathbb{E}_0(D_k \mid \mathfrak{F}_{k-1})\} = 0,$$

where iterated expectation gives $\mathbb{E}_0(D_k \mid \mathfrak{F}_{k-1}) = 0$. Hence square integrability implies

$$\mathrm{Var}_0 \left(\sum_{j=1}^{J} D_j \right) = \sum_{j=1}^{J} \mathbb{E}_0(D_j^2) = \mathbb{E}_0 \left(\sum_{j=1}^{J} w_j^2 v_j \right).$$

□

Proposition A.2 (Local expansion)**.** *For each J, let $Y_{J,1}, \ldots, Y_{J,J}$ be adapted to a filtration $(\mathfrak{H}_{J,j})_{j=0}^{J}$. Under $\mathbb{P}_0$, suppose the conditional law of $Y_{J,j}$ given $\mathfrak{H}_{J,j-1}$ is a deterministic finite-support distribution with cumulant function*

$$\psi_{J,j}(u) = \log \mathbb{E}_0 \{\exp(uY_{J,j}) \mid \mathfrak{H}_{J,j-1}\},$$

and write $\mu_{J,j} = \psi_{J,j}'(0)$ and $v_{J,j} = \psi_{J,j}''(0)$. Let $w_{J,j}$ be deterministic and define

$$V_J = \sum_{j=1}^{J} w_{J,j}^2 v_{J,j}, \qquad Z_J = V_J^{-1/2} \sum_{j=1}^{J} w_{J,j}(Y_{J,j} - \mu_{J,j}),$$

where $V_J \to \infty$ and the normalized quantities are defined for J with $V_J > 0$. Suppose, for every fixed $c > 0$,

$$V_J^{-3/2} \sum_{j=1}^{J} |w_{J,j}|^3 \sup_{|u| \le c|w_{J,j}|/\sqrt{V_J}} |\psi_{J,j}'''(u)| \longrightarrow 0.$$

For

$$E_J^{\triangle}(\eta) = \prod_{j=1}^{J} \exp\{\eta w_{J,j} Y_{J,j} - \psi_{J,j}(\eta w_{J,j})\}$$

and the prespecified local sequence $\eta_J = a/\sqrt{V_J}$, with fixed $a \in \mathbb{R}$,

$$\log E_J^{\triangle}(\eta_J) = aZ_J - a^2/2 + o(1).$$

Proof. For each j, Taylor's theorem gives

$$\psi_{J,j}(\eta_J w_{J,j}) = \psi_{J,j}(0) + \eta_J w_{J,j}\psi'_{J,j}(0) + \frac{\eta_J^2 w_{J,j}^2}{2}\psi''_{J,j}(0) + R_{J,j},$$

where

$$|R_{J,j}| \le \frac{|\eta_J w_{J,j}|^3}{6} \sup_{|u|\le|a||w_{J,j}|/\sqrt{V_J}} |\psi'''_{J,j}(u)|.$$

Since $\psi_{J,j}(0) = 0$, $\psi'_{J,j}(0) = \mu_{J,j}$ and $\psi''_{J,j}(0) = v_{J,j}$,

$$\begin{aligned}\log E_J^{\triangle}(\eta_J) &= \eta_J \sum_{j=1}^{J} w_{J,j}(Y_{J,j} - \mu_{J,j}) - \frac{\eta_J^2}{2}\sum_{j=1}^{J} w_{J,j}^2 v_{J,j} - \sum_{j=1}^{J} R_{J,j} \\ &= aZ_J - \frac{a^2}{2} - \sum_{j=1}^{J} R_{J,j}.\end{aligned}$$

If $a = 0$, the result is immediate. If $a \ne 0$, the third-derivative condition with $c = |a|$ implies $\sum_{j=1}^{J} |R_{J,j}| \to 0$. ☐

Corollary A.3 (Gaussian limit under a score central limit theorem)**.** *Under the conditions of Proposition A.2, if $Z_J \Rightarrow \mathrm{N}(0,1)$ under $\mathbb{P}_0$, then*

$$\log E_J^{\triangle}(\eta_J) \Rightarrow \mathrm{N}\left(-a^2/2, a^2\right).$$

Proof. Apply Slutsky's theorem to Proposition A.2. ☐

A.4 Sufficient conditions for exact-event updates

Let $(\mathfrak{F}_t^{\mathrm{ex}})_{t\ge0}$ be a filtration satisfying the usual conditions. Let $N_i(t)$ be a simple $\mathfrak{F}_t^{\mathrm{ex}}$-adapted observed event counting process and $R_i(t)$ an $\mathfrak{F}_t^{\mathrm{ex}}$-predictable at-risk indicator. The filtration includes the baseline covariates, group labels and previously observed information from all strata. Fix one monitoring and covariate stratum and set $N_i = R_i = 0$ for subjects outside it. This is the standard multiplicative-intensity formulation (Andersen et al., 1993; Fleming and Harrington, 1991; ter Schure et al., 2024).

Condition A.4 (Common null intensity)**.** *For every $\mathbb{P}_0 \in \mathcal{P}_0$, the multivariate counting process is nonexplosive on every finite time interval, distinct subjects have no simultaneous event jumps, and N_i has compensator*

$$\Gamma_i(t) = \int_0^t R_i(s)\lambda(s)\,\mathrm{d}s$$

for a common nonnegative $\mathfrak{F}_t^{\mathrm{ex}}$-predictable process λ, which may depend on $\mathbb{P}_0$.

Let $N_\cdot = \sum_{i=1}^{n} N_i$, let T_k^{ex} be its kth finite jump time, and let $I_k \in \{1,\dots,n\}$ be the index of the unique subject whose process jumps at T_k^{ex}. We call I_k the *mark* of this observed failure.

Proposition A.5 (Uniform mark law for an exact failure)**.** *Under Condition A.4, on the event that $T_k^{\mathrm{ex}} < \infty$ and the total risk-set intensity at T_k^{ex} is positive,*

$$\mathbb{P}_0\left(I_k = i \mid \mathfrak{F}_{T_k^{\mathrm{ex}}-}^{\mathrm{ex}} \vee \sigma(T_k^{\mathrm{ex}})\right) = \frac{R_i(T_k^{\mathrm{ex}})}{\sum_{r=1}^{n} R_r(T_k^{\mathrm{ex}})}.$$

Thus the mark is uniform over the current risk set

$$\mathsf{R}(T_k^{\mathrm{ex}}) = \{i \in \{1,\dots,n\} : R_i(T_k^{\mathrm{ex}}) = 1\}.$$

Proof. Regard the observed events as a marked point process on the product of time and the finite mark space $\{1, \ldots, n\}$. Under Condition A.4, its predictable compensator measure is

$$\nu(\mathrm{d}t, \{i\}) = R_i(t)\lambda(t)\,\mathrm{d}t,$$

with ground compensator

$$\nu_{\cdot}(\mathrm{d}t) = \lambda(t)\sum_{r=1}^{n} R_r(t)\,\mathrm{d}t.$$

Write

$$\pi_i(t) = \frac{R_i(t)}{\sum_{r=1}^{n} R_r(t)}$$

when the denominator is positive, and set $\pi_i(t) = 0$ otherwise. For any bounded nonnegative predictable H, the compensator identities, with the predictable selector $\mathbf{1}\{N_{\cdot}(t-) = k-1\}$, give

$$\begin{aligned}
&\mathbb{E}_0\left[H(T_k^{\mathrm{ex}})\mathbf{1}\{T_k^{\mathrm{ex}} < \infty, I_k = i\}\right] \\
&\quad = \mathbb{E}_0 \int_0^{\infty} H(t)\mathbf{1}\{N_{\cdot}(t-) = k-1\}R_i(t)\lambda(t)\,\mathrm{d}t \\
&\quad = \mathbb{E}_0 \int_0^{\infty} H(t)\mathbf{1}\{N_{\cdot}(t-) = k-1\}\pi_i(t)\lambda(t)\sum_{r=1}^{n} R_r(t)\,\mathrm{d}t \\
&\quad = \mathbb{E}_0\left[H(T_k^{\mathrm{ex}})\mathbf{1}\{T_k^{\mathrm{ex}} < \infty\}\pi_i(T_k^{\mathrm{ex}})\right].
\end{aligned}$$

The identities may first be applied up to bounded times and localizing times and then extended by monotone convergence. Indicators of the form $H(t) = \mathbf{1}_A\mathbf{1}\{t > s\}$, with $A \in \mathfrak{F}_s^{\mathrm{ex}}$, together with initial measurable indicators, generate $\mathfrak{F}_{T_k^{\mathrm{ex}}-}^{\mathrm{ex}}$ upon evaluation at T_k^{ex}. A monotone-class argument therefore identifies $\pi_i(T_k^{\mathrm{ex}})$ as the conditional mark probability on $\{T_k^{\mathrm{ex}} < \infty\}$. The jump time itself is measurable with respect to this pre-jump sigma-algebra. Cancelling the common positive intensity yields the asserted uniform distribution on the current stratum's risk set. □

Suppose update j corresponds to T_k^{ex}, with $\mathsf{R}_j = \mathsf{R}(T_k^{\mathrm{ex}})$. If

$$\mathfrak{G}_j \subseteq \mathfrak{F}_{T_k^{\mathrm{ex}}-}^{\mathrm{ex}} \vee \sigma(T_k^{\mathrm{ex}}),$$

then the tower property and $\mathfrak{G}_j$-measurability of the risk set give

$$\mathbb{P}_0(I_k = i \mid \mathfrak{G}_j) = \frac{\mathbf{1}\{i \in \mathsf{R}_j\}}{N_j}.$$

Summing over treated subjects gives the $M_j = 1$ specialization of (4). If $\mathfrak{G}_j$ also contains allocations processed in other strata at the same time, the uniform law must hold conditional on those allocations as well; the common-intensity condition alone does not establish this stronger statement.

When $M_j = 1$ and $w_j = 1$, the normalising constant in (7) is

$$\frac{N_{0j} + e^{\eta}N_{1j}}{N_j}.$$

The resulting e-factor is

$$e_j(\eta) = \begin{cases} \dfrac{e^{\eta}N_j}{N_{0j} + e^{\eta}N_{1j}}, & Y_j = 1, \\ \dfrac{N_j}{N_{0j} + e^{\eta}N_{1j}}, & Y_j = 0, \end{cases}$$

which is the explicit $M_j = 1$ specialization of (8).

A.5 Sufficient conditions for interval-censored and left-censored updates

For a completed interval $(\ell_j, r_j]$, let

$$D_{ij} = \mathbf{1}\{\ell_j < T_i \le r_j\}, \qquad i \in \mathsf{R}_j.$$

If $(D_{ij} : i \in \mathsf{R}_j)$ is exchangeable conditional on $\mathfrak{K}_j$, then every binary vector with the same number of ones has the same conditional probability. Since $M_j = \sum_{i \in \mathsf{R}_j} D_{ij}$, conditioning further on $M_j = m$ makes every size-m event subset equiprobable and establishes Condition 2.1.

Under the stronger independent-Bernoulli version of (5), for every $\mathsf{A} \subseteq \mathsf{R}_j$ with $|\mathsf{A}| = m$,

$$\mathbb{P}_0 \left(\mathsf{A}_j = \mathsf{A} \mid \mathfrak{K}_j\right) = p_j^m (1 - p_j)^{N_j - m}.$$

The same fixed-total conditioning therefore applies.

In the basic interval assessment model, let S denote the common stratum-specific population survival function. In the absence of additional history-dependent heterogeneity, a subject known event-free at ℓ_j has, provided $S(\ell_j) > 0$,

$$p_j = 1 - \frac{S(r_j)}{S(\ell_j)}.$$

The formula remains valid when $\lim_{t \to \infty} S(t) > 0$ because of a cure fraction. For $\ell_j = 0$, $S(0) = 1$ and $p_j = 1 - S(r_j)$.

A.6 Right censoring and completed observation windows

Let

$$\mathfrak{L}_j = \mathfrak{F}_{j-1} \vee \sigma(O_j)$$

and suppose the preselection set $\widetilde{\mathsf{R}}_j$ of subjects known event-free at ℓ_j is $\mathfrak{L}_j$-measurable. For $i \in \widetilde{\mathsf{R}}_j$, define

$$D_{ij} = \mathbf{1}\{\ell_j < T_i \le r_j\}, \qquad B_{ij} = \mathbf{1}\{\text{subject } i \text{ is observed through } r_j\},$$

and set

$$\mathsf{R}_j = \left\{ i \in \widetilde{\mathsf{R}}_j : B_{ij} = 1 \right\}.$$

Given the $\mathfrak{L}_j$-measurable set $\widetilde{\mathsf{R}}_j$, the random set R_j determines the vector $(B_{ij} : i \in \widetilde{\mathsf{R}}_j)$. Hence the pre-total sigma-algebra defined in Section 2.2 satisfies

$$\mathfrak{K}_j = \mathfrak{L}_j \vee \sigma(\mathsf{R}_j).$$

Proposition A.6 (Independent censoring preserves the interval law)**.** *Conditionally on $\mathfrak{L}_j$, suppose the pairs (D_{ij}, B_{ij}), indexed by $i \in \widetilde{\mathsf{R}}_j$, are independent across subjects, each D_{ij} is Bernoulli with a common $\mathfrak{L}_j$-measurable probability p_j, and D_{ij} is conditionally independent of B_{ij} within each pair. Then, conditionally on $\mathfrak{K}_j$, the retained indicators $(D_{ij} : i \in \mathsf{R}_j)$ are independent* Bernoulli(p_j)*. Hence Condition 2.1 holds after conditioning on M_j.*

Proof. For each subject and $d \in \{0,1\}$, conditional independence gives

$$\mathbb{P}_0\left(D_{ij} = d \mid B_{ij}, \mathfrak{L}_j\right) = \mathbb{P}_0\left(D_{ij} = d \mid \mathfrak{L}_j\right).$$

Conditional independence of the pairs implies that conditioning on the complete vector $(B_{ij} : i \in \widetilde{\mathsf{R}}_j)$ preserves the product form of the conditional law of the D_{ij}. Restricting that product to the indices with $B_{ij} = 1$ yields independent Bernoulli(p_j) indicators conditional on $\mathfrak{K}_j$. The fixed-total argument from the preceding subsection then gives the hypergeometric allocation law. ☐

A.7 Fixed-margin local score shifts

Proof of Theorem 3.3. Because the number of updates J is fixed and each $r_j(\varepsilon) = o(\varepsilon)$,

$$\mathbb{E}_\varepsilon\left\{U_J(w)\right\} = \varepsilon \sum_{j=1}^{J} w_j c_j v_j + o(\varepsilon).$$

Under $\mathbb{P}_0$, the increments $w_j(Y_j - \mu_j)$ are martingale differences for $(\mathfrak{H}_j)$ and are therefore uncorrelated. Hence

$$\mathrm{Var}_0\{U_J(w)\} = \sum_{j=1}^{J} w_j^2 v_j.$$

This proves (16). By the Cauchy–Schwarz inequality applied to the vectors $(w_j\sqrt{v_j})_{j=1}^J$ and $(c_j\sqrt{v_j})_{j=1}^J$,

$$\left(\sum_{j=1}^{J} w_j c_j v_j\right)^2 \le \left(\sum_{j=1}^{J} w_j^2 v_j\right)\left(\sum_{j=1}^{J} c_j^2 v_j\right).$$

Equality in absolute value holds exactly when w_j is proportional to c_j on the support of v_j, and the maximizing direction uses a positive proportionality constant. ☐

Proposition A.7 (Mean derivative under a local conditional-odds tilt)**.** *Fix one table with null mass function p_{0j}, mean μ_j and variance v_j. For a finite coefficient c_j, define*

$$p_{\varepsilon,j}(y) = \frac{p_{0j}(y)\exp(\varepsilon c_j y)}{\sum_{u\in\mathcal{Y}_j} p_{0j}(u)\exp(\varepsilon c_j u)}, \qquad y \in \mathcal{Y}_j.$$

Then

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\mathbb{E}_\varepsilon(Y_j)\bigg|_{\varepsilon=0} = c_j v_j,$$

so $\mathbb{E}_\varepsilon(Y_j) = \mu_j + \varepsilon c_j v_j + o(\varepsilon)$*.*

Proof. The support is finite, so differentiation may be performed term by term. If

$$\psi_j(t) = \log \sum_{u\in\mathcal{Y}_j} p_{0j}(u)e^{tu},$$

then $\mathbb{E}_\varepsilon(Y_j) = \psi_j'(\varepsilon c_j)$. Differentiating at zero gives $c_j\psi_j''(0) = c_j v_j$. ☐

For the following fixed-margin working-alternative calculations, suppose that, before conditioning on the event total, the event indicators are independent with a common probability $\pi_{gj}(\varepsilon) \in (0,1)$ within group g in the current stratum. Assume $\pi_{0j}(0) = \pi_{1j}(0)$, and write

$$\omega_j(\varepsilon) = \frac{\pi_{1j}(\varepsilon)/\{1-\pi_{1j}(\varepsilon)\}}{\pi_{0j}(\varepsilon)/\{1-\pi_{0j}(\varepsilon)\}}.$$

For fixed group sizes, the joint probability of treatment count y and total M_j is proportional, as a function of y, to

$$\binom{N_{1j}}{y}\binom{N_{0j}}{M_j - y}\omega_j(\varepsilon)^y.$$

Indeed, the remaining Bernoulli factors are $\pi_{0j}(\varepsilon)^{M_j}\{1-\pi_{0j}(\varepsilon)\}^{N_{0j}-M_j}\{1-\pi_{1j}(\varepsilon)\}^{N_{1j}}$, which do not depend on y. Conditional on the fixed total, normalization therefore gives Fisher's noncentral hypergeometric law (ter Schure et al., 2024, Section 3.3 and Appendix A.3). If $\log\omega_j(\varepsilon) = \varepsilon c_j + o(\varepsilon)$, its mean is

$$\psi_j'\{\log\omega_j(\varepsilon)\} = \mu_j + \varepsilon c_j v_j + o(\varepsilon),$$

by differentiability of the finite-support cumulant function used in Proposition A.7. This supplies (15) for the fixed-table calculation.

Proposition A.8 (Proportional-hazards local weight). *Within one prognostic stratum, let* $\Delta\Lambda_j > 0$ *be the baseline cumulative-hazard increment over a block and let the population treatment hazard ratio be* e^{ε}. *The conditional event-odds ratio is*

$$\omega_j(\varepsilon) = \frac{e^{e^{\varepsilon}\Delta\Lambda_j} - 1}{e^{\Delta\Lambda_j} - 1}.$$

Then

$$\frac{\partial}{\partial\varepsilon}\log\omega_j(\varepsilon)\bigg|_{\varepsilon=0} = \frac{\Delta\Lambda_j e^{\Delta\Lambda_j}}{e^{\Delta\Lambda_j} - 1} = 1 + \frac{\Delta\Lambda_j}{2} + O\left(\Delta\Lambda_j^2\right) \qquad \text{as } \Delta\Lambda_j \downarrow 0.$$

Proof. Differentiate

$$\log\omega_j(\varepsilon) = \log\left(e^{e^{\varepsilon}\Delta\Lambda_j} - 1\right) - \log\left(e^{\Delta\Lambda_j} - 1\right).$$

The derivative at zero is $\Delta\Lambda_j e^{\Delta\Lambda_j}/(e^{\Delta\Lambda_j} - 1)$. For $x = \Delta\Lambda_j$, divide numerator and denominator by x and use

$$\frac{xe^x}{e^x - 1} = \frac{1 + x + x^2/2 + O(x^3)}{1 + x/2 + x^2/6 + O(x^3)} = 1 + x/2 + O(x^2).$$

The denominator tends to one, so the division is valid for sufficiently small positive x. □

By Proposition A.7, this log-odds derivative is the local coefficient c_j, which approaches one as $\Delta\Lambda_j \downarrow 0$. Theorem 3.3 therefore gives approximately constant relative weights when these increments are small.

Proposition A.9 (Cure-rate local weight). *Fix* $0 \le \ell < r < \infty$. *Let* $0 \le c < 1$, $S_{\text{sus}}(\ell) > S_{\text{sus}}(r)$ *and* $S_{\text{pop},c}(r) > 0$. *At* $c = 0$ *the derivative below is understood from the right. For the mixture-cure survival function in* (17), *define the block event odds*

$$o_c(\ell, r) = \frac{S_{\text{pop},c}(\ell) - S_{\text{pop},c}(r)}{S_{\text{pop},c}(r)}.$$

Then

$$\frac{\partial}{\partial c}\log o_c(\ell, r) = -\frac{1}{(1-c)S_{\text{pop},c}(r)}.$$

Proof. Since

$$S_{\mathrm{pop},c}(\ell) - S_{\mathrm{pop},c}(r) = (1-c)\{S_{\mathrm{sus}}(\ell) - S_{\mathrm{sus}}(r)\},$$

we have

$$\log o_c(\ell, r) = \log(1-c) + \log\{S_{\mathrm{sus}}(\ell) - S_{\mathrm{sus}}(r)\} - \log S_{\mathrm{pop},c}(r).$$

Differentiate and use $\partial S_{\mathrm{pop},c}(r)/\partial c = 1 - S_{\mathrm{sus}}(r)$. Then

$$\begin{aligned}\frac{\partial}{\partial c}\log o_c(\ell, r) &= -\frac{1}{1-c} - \frac{1 - S_{\mathrm{sus}}(r)}{S_{\mathrm{pop},c}(r)} \\ &= -\frac{S_{\mathrm{pop},c}(r) + (1-c)\{1 - S_{\mathrm{sus}}(r)\}}{(1-c)S_{\mathrm{pop},c}(r)} \\ &= -\frac{1}{(1-c)S_{\mathrm{pop},c}(r)}.\end{aligned}$$

□

By Proposition A.7, the derivative supplies the local coefficient c_j up to the common scalar associated with the perturbation parameter. Within a fixed stratum its magnitude is therefore proportional to $1/S_{\mathrm{pop},c}(r)$. This is survival at the right endpoint, whereas the implemented Gray–Tsiatis rule uses the estimate preceding the update. On a consecutive common grid, ignoring estimation and truncation, the left-endpoint weight divided by the right-endpoint weight is $S_{\mathrm{pop},c}(r)/S_{\mathrm{pop},c}(\ell) = 1 - p_j$, where p_j is the conditional interval event probability. The approximation is close when p_j is small, but need not preserve relative weights across coarse intervals. The factor $1/(1-c)$ must also be retained when c varies across strata and the perturbation is a common additive change in cure probability. The fixed-margin result does not establish local optimality of the implemented weights at arbitrary assessment widths.

A.8 Confidence sequences and optional continuation

Let $\Theta \subseteq \mathbb{R}$ be a Borel parameter space and fix finite $\mathfrak{G}_j$-measurable weights w_j that do not depend on η_0. For $\eta_0 \in \Theta$, let

$$p_{\eta_0,j}(y) = \frac{p_{0j}(y)\exp(\eta_0 w_j y)}{\sum_{u\in\mathcal{Y}_j} p_{0j}(u)\exp(\eta_0 w_j u)}, \qquad y \in \mathcal{Y}_j.$$

This family models conditional event allocation, with log-odds parameter $\eta_0 w_j$ at update j. At an exact failure with $w_j = 1$, a correctly specified proportional-hazards model identifies e^{η_0} with the common treatment-to-control hazard ratio within strata. Under the independent-indicator working model in Appendix A.7, consider an assessment interval with control cumulative-hazard increment $\Delta\Lambda_j > 0$ and population hazard ratio $h > 0$. The two event probabilities are $1 - e^{-\Delta\Lambda_j}$ and $1 - e^{-h\Delta\Lambda_j}$. Their odds ratio is therefore

$$\omega_j(h) = \frac{e^{h\Delta\Lambda_j} - 1}{e^{\Delta\Lambda_j} - 1}. \tag{A.1}$$

It is generally not equal to h, although it approaches h as $\Delta\Lambda_j \downarrow 0$ for fixed h. Thus a coefficient confidence sequence for the displayed family is not automatically a hazard-ratio confidence sequence for interval-censored observations. The baseline increments would also have to be accounted for. Likewise, the derivative in the cure-rate calculation describes a perturbation at fixed susceptible survival, not a general identification of η_0 with cure probability or its group difference.

Suppose that $(\omega, \eta_0, y) \mapsto q_{j,\eta_0}(\omega, y)$ is jointly measurable, that for each η_0 it is a $\mathfrak{G}_j$-measurable probability kernel supported on the support of $p_{\eta_0,j}$, and define

$$E_J(\eta_0) = \prod_{j=1}^{J} \frac{q_{j,\eta_0}(Y_j)}{p_{\eta_0,j}(Y_j)}.$$

Theorem A.10 (Confidence sequence)**.** *For $\alpha \in (0,1)$, define the random set*

$$\mathcal{C}_J = \left\{\eta_0 \in \Theta : E_J(\eta_0) < \alpha^{-1}\right\}.$$

If, for some $\eta_\star \in \Theta$, the conditional law of Y_j given $\mathfrak{G}_j$ is $p_{\eta_\star,j}$ at every update, then

$$\mathbb{P}_{\eta_\star}\left(\eta_\star \in \mathcal{C}_J \text{ for every } J \geq 0\right) \geq 1 - \alpha.$$

Proof. At $\eta_\star$, each ratio has conditional expectation one given $\mathfrak{G}_j$, so $(E_J(\eta_\star), \mathfrak{F}_J)$ is a test martingale. Exclusion of $\eta_\star$ at any time implies $\sup_{J \geq 0} E_J(\eta_\star) \geq \alpha^{-1}$. Ville's inequality bounds the probability of this event by α. □

Theorem A.11 (Optional continuation across studies)**.** *Let $(\mathfrak{H}_j)_{j\geq 0}$ be the filtration generated by the completed studies and their initiation decisions. Suppose study j is initiated using only $\mathfrak{H}_{j-1}$ and produces an $\mathfrak{H}_j$-measurable nonnegative e-value $E^{(j)}$ satisfying*

$$\mathbb{E}_0\left(E^{(j)} \mid \mathfrak{H}_{j-1}\right) \leq 1.$$

With $K_0 = 1$, the process $K_J = \prod_{j=1}^{J} E^{(j)}$ is a nonnegative supermartingale in J. Consequently, for every $\alpha \in (0,1)$,

$$\mathbb{P}_0\left(\sup_{J \geq 0} K_J \geq \alpha^{-1}\right) \leq \alpha.$$

Proof. We have

$$\mathbb{E}_0\left(K_J \mid \mathfrak{H}_{J-1}\right) = K_{J-1}\mathbb{E}_0\left(E^{(J)} \mid \mathfrak{H}_{J-1}\right) \leq K_{J-1}.$$

Ville's inequality applies to the resulting supermartingale. □

B Additional simulation and numerical results

The supplementary studies use the sample-size grid in Section 4. The common event and observation models are in Appendix B.5.1, with scenario parameters in Table B.8; Appendix B.5 defines the procedures and performance measures. The experiment map links each specification to its results. The accompanying result files contain calibration curves, paired differences, observation proportions and Monte Carlo standard errors.

Experiment map for Appendix B.

Experiment	Specification	Results
Fixed weights, mixtures and PQ	Section 4.1; Appendix B.4	Power: Table B.6; mean capped duration: Table B.7
Alternative-specific weights	Appendix B.5.5	Power: Table B.11; interval duration and log growth: Table B.1
Primary S1/S2	Appendix B.5.1	Rejection probabilities and duration: Table B.9
Reference and effect-specific coefficients	Appendix B.5.8	Reference power, duration and growth: Table B.16; coefficient comparison: Table B.2
Mixed observations	Appendix B.3	Null: Table B.3; power: Table B.4
Interval-censored benchmark	Appendix B.5.4	Full grid: Table B.5
Allocation and censoring	Appendix B.5.7	Rejection probabilities: Tables B.13 and B.14
Assessment frequency	Appendix B.5.8	Power and conditional median: Table B.15
Covariates and delayed entry	Appendix B.5.9	Rejection probabilities: Table B.17

B.1 Weight-specific stopping and log-growth summaries

Corresponding exact-observation stopping and log-growth summaries are included in the computational supplement.

Table B.1: Alternative-specific weight specifications: interval-censored stopping and terminal log-growth summaries corresponding to Table B.11. Median stop is conditional on rejection; a dash means none rejected. Mean capped stop assigns the full planned duration to every nonrejecting replication.

n	Scenario	Weight	Median stop	Mean capped stop	Mean terminal $\log E$
100	S1	LR	5.00	7.83	0.876
100	S1	GB	3.00	7.78	0.842
100	S1	PP	4.00	7.77	0.884
100	S1	FH	7.00	8.00	0.400
100	S1	GT	6.00	7.93	0.744
100	S1	LR–GT	6.00	7.89	0.827
100	S2	LR	6.00	9.53	1.048
100	S2	GB	4.00	9.84	0.685
100	S2	PP	4.00	9.75	0.817
100	S2	FH	7.00	9.77	0.971
100	S2	GT	7.00	9.55	1.144
100	S2	LR–GT	6.00	9.53	1.132
100	S3	LR	5.50	6.00	0.410
100	S3	GB	3.00	5.96	0.588
100	S3	PP	3.50	5.96	0.554
100	S3	FH	–	6.00	0.059
100	S3	GT	–	6.00	0.219
100	S3	LR–GT	6.00	6.00	0.332
100	S4	LR	–	8.00	0.321
100	S4	GB	–	8.00	0.090
100	S4	PP	–	8.00	0.130
100	S4	FH	7.00	7.97	0.688
100	S4	GT	7.50	7.99	0.557

Continued on next page

Table B.1 — continued

n	Scenario	Weight	Median stop	Mean capped stop	Mean terminal $\log E$
100	S4	LR–GT	7.50	8.00	0.467
100	S5	LR	6.00	9.75	0.873
100	S5	GB	5.00	9.77	0.795
100	S5	PP	5.00	9.77	0.818
100	S5	FH	9.00	9.98	0.535
100	S5	GT	8.00	9.90	0.769
100	S5	LR–GT	7.00	9.83	0.844
200	S1	LR	4.00	6.75	1.756
200	S1	GB	3.00	6.68	1.678
200	S1	PP	3.00	6.64	1.767
200	S1	FH	7.00	7.89	0.825
200	S1	GT	5.00	7.20	1.503
200	S1	LR–GT	4.00	6.96	1.668
200	S2	LR	5.00	8.13	2.154
200	S2	GB	3.00	8.80	1.412
200	S2	PP	4.00	8.52	1.683
200	S2	FH	6.00	8.74	1.977
200	S2	GT	5.00	8.00	2.352
200	S2	LR–GT	5.00	8.04	2.323
200	S3	LR	3.00	5.85	0.776
200	S3	GB	2.50	5.40	1.138
200	S3	PP	2.50	5.42	1.066
200	S3	FH	–	6.00	0.103
200	S3	GT	5.50	6.00	0.404
200	S3	LR–GT	3.50	5.94	0.631
200	S4	LR	6.50	7.97	0.642
200	S4	GB	–	8.00	0.167
200	S4	PP	–	8.00	0.248
200	S4	FH	6.50	7.71	1.421
200	S4	GT	7.00	7.84	1.145
200	S4	LR–GT	7.00	7.91	0.966
200	S5	LR	5.00	8.41	1.795
200	S5	GB	4.00	8.55	1.606
200	S5	PP	4.00	8.52	1.659
200	S5	FH	8.00	9.70	1.151
200	S5	GT	6.00	8.87	1.612
200	S5	LR–GT	6.00	8.61	1.751
500	S1	LR	2.00	3.83	4.446
500	S1	GB	2.00	4.04	4.259
500	S1	PP	2.00	3.92	4.485
500	S1	FH	5.00	6.70	2.082
500	S1	GT	3.00	4.15	3.788
500	S1	LR–GT	2.00	3.94	4.245
500	S2	LR	3.00	5.05	5.490
500	S2	GB	3.00	5.95	3.593
500	S2	PP	3.00	5.61	4.293
500	S2	FH	5.00	5.98	5.051
500	S2	GT	4.00	4.83	6.014
500	S2	LR–GT	3.00	4.87	5.915
500	S3	LR	2.00	3.75	2.032
500	S3	GB	1.50	3.16	2.941
500	S3	PP	1.50	3.16	2.765
500	S3	FH	–	6.00	0.295
500	S3	GT	3.00	5.44	1.086
500	S3	LR–GT	2.50	4.28	1.717
500	S4	LR	6.50	7.53	1.756
500	S4	GB	6.00	7.99	0.495
500	S4	PP	6.50	7.94	0.715
500	S4	FH	6.00	6.55	3.736
500	S4	GT	6.00	6.95	3.052
500	S4	LR–GT	6.50	7.15	2.675
500	S5	LR	3.00	5.01	4.535
500	S5	GB	3.00	5.36	4.105

Continued on next page

Table B.1 — continued

n	Scenario	Weight	Median stop	Mean capped stop	Mean terminal $\log E$
500	S5	PP	3.00	5.28	4.234
500	S5	FH	6.00	7.62	2.843
500	S5	GT	4.00	5.35	4.028
500	S5	LR–GT	3.00	5.11	4.410
1000	S1	LR	2.00	2.06	8.884
1000	S1	GB	1.00	2.31	8.460
1000	S1	PP	1.00	2.21	8.934
1000	S1	FH	4.00	4.60	4.235
1000	S1	GT	2.00	2.09	7.614
1000	S1	LR–GT	2.00	2.04	8.559
1000	S2	LR	2.00	3.02	11.101
1000	S2	GB	2.00	3.58	7.284
1000	S2	PP	2.00	3.43	8.694
1000	S2	FH	3.00	3.96	10.182
1000	S2	GT	2.00	2.77	12.137
1000	S2	LR–GT	2.00	2.84	11.982
1000	S3	LR	1.50	1.71	4.084
1000	S3	GB	1.00	1.54	5.929
1000	S3	PP	1.00	1.53	5.575
1000	S3	FH	5.50	5.99	0.587
1000	S3	GT	2.00	2.78	2.172
1000	S3	LR–GT	1.50	1.94	3.590
1000	S4	LR	6.00	6.80	3.438
1000	S4	GB	5.50	7.83	0.936
1000	S4	PP	6.00	7.63	1.366
1000	S4	FH	5.00	5.49	7.441
1000	S4	GT	5.50	6.01	6.056
1000	S4	LR–GT	6.00	6.21	5.502
1000	S5	LR	2.00	2.74	9.212
1000	S5	GB	2.00	3.09	8.322
1000	S5	PP	2.00	3.02	8.585
1000	S5	FH	4.00	4.96	5.807
1000	S5	GT	2.00	2.87	8.203
1000	S5	LR–GT	2.00	2.73	8.991
2000	S1	LR	1.00	1.24	17.760
2000	S1	GB	1.00	1.36	17.016
2000	S1	PP	1.00	1.32	17.930
2000	S1	FH	3.00	2.92	8.338
2000	S1	GT	1.00	1.19	15.148
2000	S1	LR–GT	1.00	1.20	17.250
2000	S2	LR	2.00	1.95	22.110
2000	S2	GB	2.00	2.20	14.492
2000	S2	PP	2.00	2.19	17.304
2000	S2	FH	3.00	2.66	20.307
2000	S2	GT	1.00	1.68	24.184
2000	S2	LR–GT	2.00	1.77	23.919
2000	S3	LR	1.00	0.84	8.199
2000	S3	GB	0.50	0.80	11.887
2000	S3	PP	0.50	0.79	11.182
2000	S3	FH	4.00	5.53	1.189
2000	S3	GT	1.00	1.25	4.377
2000	S3	LR–GT	1.00	0.93	7.558
2000	S4	LR	5.50	6.00	6.833
2000	S4	GB	6.00	7.35	1.848
2000	S4	PP	6.00	7.00	2.702
2000	S4	FH	4.50	4.72	14.836
2000	S4	GT	5.00	5.22	12.083
2000	S4	LR–GT	5.00	5.36	11.414
2000	S5	LR	1.00	1.57	18.150
2000	S5	GB	1.00	1.82	16.361
2000	S5	PP	1.00	1.78	16.888
2000	S5	FH	3.00	3.26	11.510
2000	S5	GT	1.00	1.53	16.207

Continued on next page

Table B.1 — continued

n	Scenario	Weight	Median stop	Mean capped stop	Mean terminal $\log E$
2000	S5	LR–GT	1.00	1.52	17.799

B.2 Effect-specific log-growth calibration

The fixed coefficient in Table B.16 maximizes pilot mean terminal log evidence at the $n = 500$ middle design alternative. Here a separate independent pilot of 1,000 alternative datasets at each evaluation sample size and true effect selects an effect-specific coefficient for the same objective on the same 50-point grid. The two coefficients are evaluated on the same alternative datasets. Maximizing pilot mean terminal log evidence need not maximize the probability of crossing by the planned end of follow-up, especially under weak effects.

Table B.2: Fixed middle-design (D) versus effect-specific (S) coefficients. S1 and S2 denote the latency and cure families in Appendix B.5.8, with susceptible HR and treatment cure probability as the respective effects. These effects vary from the primary specifications in Table B.8. Growth is mean terminal log e-value.

n	Family	Effect	Obs.	η_D	η_S	Power D	Power S	Growth D	Growth S
100	S1	0.60	Interval	-0.343	-0.494	0.050	0.163	1.085	1.176
100	S1	0.60	Exact	-0.317	-0.469	0.052	0.190	1.082	1.179
100	S1	0.70	Interval	-0.343	-0.317	0.022	0.013	0.610	0.608
100	S1	0.70	Exact	-0.317	-0.292	0.025	0.015	0.612	0.609
100	S1	0.80	Interval	-0.343	-0.217	0.009	0.000	0.157	0.251
100	S1	0.80	Exact	-0.317	-0.191	0.011	0.000	0.161	0.251
100	S2	0.25	Interval	-0.217	-0.116	0.041	0.001	0.060	0.335
100	S2	0.25	Exact	-0.217	-0.116	0.045	0.001	0.068	0.333
100	S2	0.34	Interval	-0.217	-0.217	0.146	0.146	1.179	1.179
100	S2	0.34	Exact	-0.217	-0.217	0.156	0.156	1.176	1.176
100	S2	0.43	Interval	-0.217	-0.317	0.332	0.478	2.183	2.442
100	S2	0.43	Exact	-0.217	-0.343	0.331	0.519	2.156	2.435
200	S1	0.60	Interval	-0.343	-0.494	0.380	0.512	2.253	2.462
200	S1	0.60	Exact	-0.317	-0.469	0.416	0.564	2.243	2.465
200	S1	0.70	Interval	-0.343	-0.317	0.171	0.149	1.175	1.176
200	S1	0.70	Exact	-0.317	-0.292	0.194	0.164	1.182	1.180
200	S1	0.80	Interval	-0.343	-0.217	0.075	0.009	0.295	0.492
200	S1	0.80	Exact	-0.317	-0.191	0.095	0.009	0.303	0.495
200	S2	0.25	Interval	-0.217	-0.116	0.149	0.035	0.174	0.707
200	S2	0.25	Exact	-0.217	-0.116	0.173	0.038	0.194	0.707
200	S2	0.34	Interval	-0.217	-0.217	0.469	0.469	2.358	2.358
200	S2	0.34	Exact	-0.217	-0.217	0.499	0.499	2.356	2.356
200	S2	0.43	Interval	-0.217	-0.317	0.826	0.839	4.457	5.000
200	S2	0.43	Exact	-0.217	-0.317	0.840	0.865	4.403	4.987
500	S1	0.60	Interval	-0.343	-0.494	0.900	0.886	5.580	6.072
500	S1	0.60	Exact	-0.317	-0.443	0.922	0.920	5.566	6.098
500	S1	0.70	Interval	-0.343	-0.343	0.621	0.621	3.065	3.065
500	S1	0.70	Exact	-0.317	-0.317	0.685	0.685	3.076	3.076
500	S1	0.80	Interval	-0.343	-0.217	0.287	0.193	0.742	1.238
500	S1	0.80	Exact	-0.317	-0.191	0.349	0.211	0.763	1.246
500	S2	0.25	Interval	-0.217	-0.116	0.355	0.320	0.392	1.766
500	S2	0.25	Exact	-0.217	-0.116	0.429	0.354	0.445	1.765
500	S2	0.34	Interval	-0.217	-0.217	0.890	0.890	5.996	5.996
500	S2	0.34	Exact	-0.217	-0.217	0.914	0.914	5.992	5.992
500	S2	0.43	Interval	-0.217	-0.317	0.996	0.989	11.136	12.447
500	S2	0.43	Exact	-0.217	-0.317	0.998	0.993	11.011	12.429
1000	S1	0.60	Interval	-0.343	-0.469	0.997	0.991	11.249	12.261
1000	S1	0.60	Exact	-0.317	-0.443	0.999	0.995	11.222	12.313
1000	S1	0.70	Interval	-0.343	-0.343	0.895	0.895	6.154	6.154
1000	S1	0.70	Exact	-0.317	-0.317	0.930	0.930	6.186	6.186
1000	S1	0.80	Interval	-0.343	-0.217	0.494	0.519	1.429	2.447
1000	S1	0.80	Exact	-0.317	-0.191	0.593	0.572	1.470	2.464
1000	S2	0.25	Interval	-0.217	-0.116	0.519	0.672	0.830	3.569
1000	S2	0.25	Exact	-0.217	-0.116	0.616	0.719	0.938	3.568
1000	S2	0.34	Interval	-0.217	-0.217	0.987	0.987	12.172	12.172
1000	S2	0.34	Exact	-0.217	-0.217	0.991	0.991	12.166	12.166
1000	S2	0.43	Interval	-0.217	-0.317	1.000	1.000	22.304	24.898
1000	S2	0.43	Exact	-0.217	-0.343	1.000	1.000	22.057	24.847
2000	S1	0.60	Interval	-0.343	-0.494	1.000	1.000	22.519	24.551
2000	S1	0.60	Exact	-0.317	-0.443	1.000	1.000	22.460	24.645
2000	S1	0.70	Interval	-0.343	-0.343	0.991	0.991	12.440	12.440
2000	S1	0.70	Exact	-0.317	-0.317	0.995	0.995	12.501	12.501
2000	S1	0.80	Interval	-0.343	-0.217	0.687	0.825	2.861	4.898
2000	S1	0.80	Exact	-0.317	-0.191	0.796	0.877	2.949	4.935
2000	S2	0.25	Interval	-0.217	-0.116	0.658	0.922	1.737	7.191
2000	S2	0.25	Exact	-0.217	-0.116	0.765	0.948	1.955	7.191
2000	S2	0.34	Interval	-0.217	-0.217	1.000	1.000	23.894	23.894
2000	S2	0.34	Exact	-0.217	-0.217	1.000	1.000	23.886	23.886

Continued on next page

Table B.2 — continued

n	Family	Effect	Obs.	η_D	η_S	Power D	Power S	Growth D	Growth S
2000	S2	0.43	Interval	-0.217	-0.317	1.000	1.000	44.966	50.315
2000	S2	0.43	Exact	-0.217	-0.317	1.000	1.000	44.467	50.244

B.3 Mixed observation schemes

For the design in Section 4.5, write $r_{\text{exact}} \in \{0, 0.25, 0.5, 0.75, 1\}$ for the exact-observation fraction. Within each replicate, subjects whose latent uniform draw is below the selected fraction are observed exactly and the remainder annually. Exact subcohort uses only those observed exactly; the all-interval and all-exact evaluation benchmarks are shared across fractions.

Table B.3: Null crossing probability of the unified process by total sample size and exact-observation fraction.

n	Scenario	0	0.25	0.50	0.75	1
100	S1	0.0050	0.0044	0.0050	0.0054	0.0066
100	S2	0.0074	0.0080	0.0074	0.0126	0.0122
200	S1	0.0106	0.0124	0.0150	0.0154	0.0160
200	S2	0.0144	0.0174	0.0198	0.0238	0.0224
500	S1	0.0222	0.0298	0.0336	0.0384	0.0388
500	S2	0.0230	0.0322	0.0360	0.0396	0.0402
1000	S1	0.0144	0.0232	0.0340	0.0440	0.0428
1000	S2	0.0178	0.0312	0.0378	0.0448	0.0464
2000	S1	0.0056	0.0202	0.0364	0.0424	0.0464
2000	S2	0.0120	0.0282	0.0400	0.0480	0.0516

Table B.4: Power by the planned end of follow-up, by sample size and exact-observation fraction. All exact uses additional event-time information.

n	Scenario	Analysis	0	0.25	0.50	0.75	1
100	S1	Unified	0.0792	0.0678	0.0744	0.0822	0.0952
100	S1	All interval	0.0792	0.0792	0.0792	0.0792	0.0792
100	S1	Exact subcohort	0.0000	0.0000	0.0054	0.0368	0.0952
100	S1	All exact (benchmark)	0.0952	0.0952	0.0952	0.0952	0.0952
100	S2	Unified	0.1434	0.1346	0.1318	0.1762	0.1862
100	S2	All interval	0.1434	0.1434	0.1434	0.1434	0.1434
100	S2	Exact subcohort	0.0000	0.0000	0.0216	0.0918	0.1862
100	S2	All exact (benchmark)	0.1862	0.1862	0.1862	0.1862	0.1862
200	S1	Unified	0.3448	0.3420	0.3630	0.3756	0.3920
200	S1	All interval	0.3448	0.3448	0.3448	0.3448	0.3448
200	S1	Exact subcohort	0.0000	0.0060	0.0980	0.2462	0.3920
200	S1	All exact (benchmark)	0.3920	0.3920	0.3920	0.3920	0.3920
200	S2	Unified	0.4710	0.4752	0.4866	0.4916	0.5040
200	S2	All interval	0.4710	0.4710	0.4710	0.4710	0.4710
200	S2	Exact subcohort	0.0000	0.0104	0.1500	0.3390	0.5040
200	S2	All exact (benchmark)	0.5040	0.5040	0.5040	0.5040	0.5040
500	S1	Unified	0.7856	0.8038	0.8190	0.8244	0.8290
500	S1	All interval	0.7856	0.7856	0.7856	0.7856	0.7856
500	S1	Exact subcohort	0.0000	0.1376	0.4924	0.7054	0.8290
500	S1	All exact (benchmark)	0.8290	0.8290	0.8290	0.8290	0.8290
500	S2	Unified	0.8776	0.8892	0.8984	0.9040	0.9090
500	S2	All interval	0.8776	0.8776	0.8776	0.8776	0.8776
500	S2	Exact subcohort	0.0000	0.2436	0.6366	0.8142	0.9090
500	S2	All exact (benchmark)	0.9090	0.9090	0.9090	0.9090	0.9090
1000	S1	Unified	0.9682	0.9730	0.9752	0.9748	0.9790

Continued on next page

Table B.4 — continued

n	Scenario	Analysis	0	0.25	0.50	0.75	1
1000	S1	All interval	0.9682	0.9682	0.9682	0.9682	0.9682
1000	S1	Exact subcohort	0.0000	0.4886	0.8386	0.9342	0.9790
1000	S1	All exact (benchmark)	0.9790	0.9790	0.9790	0.9790	0.9790
1000	S2	Unified	0.9870	0.9886	0.9902	0.9906	0.9906
1000	S2	All interval	0.9870	0.9870	0.9870	0.9870	0.9870
1000	S2	Exact subcohort	0.0000	0.6256	0.9074	0.9732	0.9906
1000	S2	All exact (benchmark)	0.9906	0.9906	0.9906	0.9906	0.9906
2000	S1	Unified	0.9978	0.9996	0.9992	0.9996	0.9994
2000	S1	All interval	0.9978	0.9978	0.9978	0.9978	0.9978
2000	S1	Exact subcohort	0.0000	0.8306	0.9706	0.9954	0.9994
2000	S1	All exact (benchmark)	0.9994	0.9994	0.9994	0.9994	0.9994
2000	S2	Unified	0.9996	0.9996	0.9998	0.9998	0.9998
2000	S2	All interval	0.9996	0.9996	0.9996	0.9996	0.9996
2000	S2	Exact subcohort	0.0000	0.9090	0.9900	0.9986	0.9998
2000	S2	All exact (benchmark)	0.9998	0.9998	0.9998	0.9998	0.9998

B.4 Additional benchmark and fixed-design comparisons

For the fixed-design study, the reference-null normalization multipliers for LR, GB, PP, FH and GT are approximately $(1, 1.35180, 1.27465, 3.15151, 0.65365)$ for interval observation and $(1, 1.45782, 1.37477, 2.84439, 0.61015)$ for exact observation. Full-precision constants, coefficients, seeds, binomial intervals and paired Monte Carlo uncertainties are retained with the reproducibility materials.

For the prequential comparators, let a and b be the initial enrolled control and treatment counts, respectively. When both are positive, the smoothing factor of ter Schure et al. (2024, Section 3.2, equation (3.7)) is

$$s(u) = \frac{e^u}{\{a + 1 + (b + 1)e^u\}\{a + (b + 1)e^u\}}.$$

Let t_j be the release time of comparison j, and let $q^{(1)}_{u,j}$ denote (7) with unit weight. Before a release at time t, fit

$$\widehat{\eta}(t) = \underset{u \in \mathbb{R}}{\arg\max} \left\{ \log s(u) + \sum_{j : t_j < t} \log q^{(1)}_{u,j}(Y_j) \right\}.$$

For exact failures this is the published smoothed fit; for grouped observations the sum uses the conditional likelihoods of the earlier completed-window tables. Before any informative comparison, the fit is $\log\{\sqrt{a(a + 1)/(b + 1)}\}$. The smoothing gives a unique finite maximizer; no parameter bounds are imposed. All comparisons released together use the same fit, and enter its history only after their evidence factors have been computed. PQ uses $\widehat{\eta}(t)$ and PQ$^-$ uses $\min\{\widehat{\eta}(t), 0\}$ in the unit-weight factor. If either enrolled group is empty, both processes remain one. The supplement implements these rules in `sim2_comparators` in `scripts/sim2_comparators.R`, with the likelihood and fitting calculation in `scripts/sim2_kernels.cpp`.

Table B.5: Fixed-analysis interval-censored benchmark over the full sample-size grid. Δ is Fay–Shaw power minus Fixed LR power, with its paired Monte Carlo SE.

n	Scenario	Fixed LR		Fay–Shaw		Δ	Paired SE
		Null	Power	Null	Power		
100	S1	0.0524	0.3866	0.0520	0.3846	-0.0020	0.0008
100	S2	0.0488	0.4392	0.0490	0.4374	-0.0018	0.0008
100	S3	0.0544	0.2302	0.0528	0.2276	-0.0026	0.0011
100	S4	0.0530	0.2090	0.0520	0.2070	-0.0020	0.0007
100	S5	0.0554	0.3908	0.0538	0.3890	-0.0018	0.0011
200	S1	0.0502	0.5982	0.0496	0.5972	-0.0010	0.0005
200	S2	0.0446	0.6730	0.0452	0.6720	-0.0010	0.0007
200	S3	0.0484	0.3562	0.0480	0.3548	-0.0014	0.0007
200	S4	0.0506	0.3266	0.0508	0.3258	-0.0008	0.0007
200	S5	0.0456	0.5998	0.0452	0.5982	-0.0016	0.0008
500	S1	0.0462	0.9030	0.0462	0.9026	-0.0004	0.0003
500	S2	0.0476	0.9542	0.0476	0.9540	-0.0002	0.0002
500	S3	0.0512	0.6518	0.0512	0.6516	-0.0002	0.0004
500	S4	0.0478	0.5956	0.0480	0.5954	-0.0002	0.0003
500	S5	0.0492	0.9234	0.0492	0.9228	-0.0006	0.0003
1000	S1	0.0496	0.9946	0.0496	0.9946	0.0000	0.0000
1000	S2	0.0448	0.9992	0.0448	0.9992	0.0000	0.0000
1000	S3	0.0552	0.8894	0.0548	0.8890	-0.0004	0.0004
1000	S4	0.0474	0.8398	0.0472	0.8396	-0.0002	0.0002
1000	S5	0.0464	0.9962	0.0464	0.9962	0.0000	0.0000
2000	S1	0.0530	0.9998	0.0530	0.9998	0.0000	0.0000
2000	S2	0.0510	1.0000	0.0508	1.0000	0.0000	0.0000
2000	S3	0.0498	0.9926	0.0498	0.9926	0.0000	0.0000
2000	S4	0.0482	0.9820	0.0484	0.9820	0.0000	0.0000
2000	S5	0.0498	1.0000	0.0498	1.0000	0.0000	0.0000

Table B.6: Power for fixed practical designs and prequential comparators over the full sample-size grid.

n	Scenario	Obs.	LR	GB	PP	FH	GT	Cmix	Wmix	PQ	PQ^{-}
100	S1	Interval	0.049	0.042	0.051	0.009	0.027	0.105	0.074	0.067	0.074
100	S1	Exact	0.070	0.063	0.074	0.023	0.044	0.134	0.107	0.117	0.135
100	S2	Interval	0.063	0.035	0.045	0.060	0.081	0.114	0.130	0.073	0.083
100	S2	Exact	0.085	0.052	0.061	0.092	0.104	0.133	0.155	0.090	0.107
100	S3	Interval	0.035	0.039	0.040	0.009	0.021	0.082	0.054	0.067	0.072
100	S3	Exact	0.051	0.064	0.062	0.016	0.029	0.104	0.075	0.108	0.119
100	S4	Interval	0.022	0.007	0.009	0.081	0.056	0.037	0.070	0.022	0.023
100	S4	Exact	0.027	0.012	0.013	0.077	0.060	0.040	0.075	0.034	0.033
100	S5	Interval	0.064	0.051	0.049	0.048	0.069	0.116	0.109	0.072	0.081
100	S5	Exact	0.093	0.075	0.072	0.077	0.091	0.143	0.145	0.099	0.116
200	S1	Interval	0.280	0.234	0.263	0.090	0.223	0.290	0.260	0.148	0.166
200	S1	Exact	0.346	0.288	0.328	0.193	0.296	0.352	0.345	0.223	0.263
200	S2	Interval	0.332	0.193	0.243	0.329	0.398	0.300	0.358	0.198	0.218
200	S2	Exact	0.368	0.229	0.277	0.418	0.441	0.330	0.402	0.195	0.237
200	S3	Interval	0.227	0.235	0.236	0.052	0.150	0.248	0.200	0.139	0.156
200	S3	Exact	0.287	0.291	0.295	0.097	0.206	0.306	0.266	0.211	0.243
200	S4	Interval	0.101	0.026	0.038	0.277	0.219	0.073	0.167	0.044	0.047
200	S4	Exact	0.112	0.034	0.047	0.284	0.238	0.080	0.181	0.051	0.052
200	S5	Interval	0.323	0.258	0.265	0.231	0.325	0.296	0.298	0.171	0.192
200	S5	Exact	0.375	0.305	0.314	0.331	0.388	0.342	0.367	0.212	0.247
500	S1	Interval	0.775	0.690	0.734	0.500	0.765	0.677	0.676	0.466	0.507
500	S1	Exact	0.821	0.729	0.774	0.725	0.832	0.739	0.767	0.576	0.633
500	S2	Interval	0.823	0.593	0.689	0.834	0.890	0.725	0.797	0.608	0.647
500	S2	Exact	0.826	0.600	0.693	0.891	0.906	0.740	0.834	0.540	0.626
500	S3	Interval	0.708	0.684	0.697	0.275	0.651	0.611	0.590	0.413	0.442
500	S3	Exact	0.780	0.734	0.751	0.497	0.748	0.688	0.696	0.503	0.566
500	S4	Interval	0.285	0.077	0.111	0.639	0.528	0.191	0.443	0.137	0.162
500	S4	Exact	0.294	0.085	0.115	0.653	0.557	0.204	0.456	0.105	0.150
500	S5	Interval	0.777	0.680	0.706	0.650	0.806	0.675	0.685	0.508	0.547
500	S5	Exact	0.807	0.701	0.727	0.800	0.860	0.721	0.767	0.523	0.602
1000	S1	Interval	0.972	0.945	0.961	0.854	0.978	0.954	0.952	0.802	0.835
1000	S1	Exact	0.981	0.952	0.966	0.969	0.988	0.968	0.975	0.903	0.932
1000	S2	Interval	0.980	0.868	0.931	0.981	0.993	0.969	0.985	0.932	0.949
1000	S2	Exact	0.976	0.854	0.919	0.993	0.994	0.968	0.988	0.912	0.950
1000	S3	Interval	0.944	0.920	0.930	0.720	0.950	0.900	0.895	0.745	0.773
1000	S3	Exact	0.967	0.936	0.944	0.906	0.976	0.936	0.945	0.830	0.876
1000	S4	Interval	0.429	0.089	0.143	0.846	0.743	0.362	0.763	0.325	0.397
1000	S4	Exact	0.417	0.095	0.144	0.856	0.762	0.364	0.763	0.236	0.359
1000	S5	Interval	0.959	0.913	0.931	0.898	0.970	0.936	0.940	0.839	0.865
1000	S5	Exact	0.966	0.914	0.933	0.970	0.981	0.949	0.967	0.878	0.914
2000	S1	Interval	0.999	0.996	0.997	0.990	1.000	0.999	0.999	0.965	0.976
2000	S1	Exact	0.999	0.995	0.997	0.999	1.000	0.999	1.000	0.998	0.999
2000	S2	Interval	1.000	0.982	0.995	1.000	1.000	1.000	1.000	0.999	0.999
2000	S2	Exact	0.999	0.974	0.993	1.000	1.000	1.000	1.000	0.999	1.000
2000	S3	Interval	0.998	0.992	0.995	0.984	1.000	0.996	0.998	0.963	0.973
2000	S3	Exact	0.999	0.994	0.996	0.999	1.000	0.999	1.000	0.993	0.997
2000	S4	Interval	0.574	0.085	0.156	0.953	0.889	0.708	0.982	0.670	0.794
2000	S4	Exact	0.539	0.095	0.153	0.957	0.899	0.730	0.981	0.565	0.761
2000	S5	Interval	0.998	0.992	0.995	0.991	1.000	1.000	0.999	0.986	0.991
2000	S5	Exact	0.998	0.988	0.993	0.999	1.000	1.000	1.000	0.997	1.000

Table B.7: Mean capped stopping time in calendar years for fixed practical designs and prequential comparators over the full sample-size grid.

n	Scenario	Obs.	LR	GB	PP	FH	GT	Cmix	Wmix	PQ	PQ^{-}
100	S1	Interval	7.88	7.82	7.82	7.99	7.96	7.61	7.79	7.76	7.73
100	S1	Exact	7.75	7.67	7.66	7.96	7.90	7.39	7.59	7.34	7.25
100	S2	Interval	9.74	9.79	9.77	9.86	9.78	9.46	9.51	9.65	9.60
100	S2	Exact	9.57	9.65	9.63	9.70	9.61	9.24	9.26	9.36	9.26
100	S3	Interval	5.92	5.88	5.88	5.99	5.98	5.75	5.87	5.77	5.75
100	S3	Exact	5.86	5.77	5.79	5.98	5.95	5.64	5.79	5.53	5.48
100	S4	Interval	7.96	7.98	7.97	7.90	7.93	7.91	7.87	7.93	7.93
100	S4	Exact	7.94	7.96	7.95	7.88	7.90	7.88	7.83	7.83	7.86
100	S5	Interval	9.77	9.73	9.76	9.91	9.84	9.48	9.60	9.67	9.63
100	S5	Exact	9.58	9.55	9.59	9.78	9.70	9.22	9.34	9.30	9.20
200	S1	Interval	6.97	6.87	6.83	7.86	7.41	6.74	7.04	7.45	7.37
200	S1	Exact	6.41	6.36	6.27	7.46	6.96	6.15	6.43	6.71	6.49
200	S2	Interval	8.42	8.79	8.64	8.98	8.50	8.42	8.37	9.11	9.01
200	S2	Exact	7.88	8.34	8.16	8.17	7.90	7.90	7.74	8.72	8.45
200	S3	Interval	5.27	5.12	5.15	5.92	5.63	5.07	5.33	5.50	5.44
200	S3	Exact	4.96	4.78	4.81	5.80	5.43	4.74	5.00	5.04	4.90
200	S4	Interval	7.79	7.91	7.89	7.50	7.63	7.83	7.67	7.89	7.90
200	S4	Exact	7.71	7.86	7.83	7.41	7.53	7.76	7.57	7.76	7.81
200	S5	Interval	8.48	8.49	8.53	9.33	8.78	8.41	8.60	9.21	9.10
200	S5	Exact	7.84	7.91	7.94	8.66	8.20	7.79	7.87	8.56	8.31
500	S1	Interval	4.00	4.10	3.99	6.52	4.47	4.49	4.68	6.04	5.83
500	S1	Exact	3.08	3.27	3.13	4.81	3.48	3.50	3.56	4.67	4.27
500	S2	Interval	5.00	5.95	5.57	6.08	4.94	5.88	5.63	7.08	6.82
500	S2	Exact	4.18	5.31	4.88	4.47	3.96	5.02	4.48	6.63	6.00
500	S3	Interval	3.02	2.98	2.96	5.18	3.47	3.36	3.54	4.36	4.24
500	S3	Exact	2.43	2.46	2.42	4.27	2.87	2.75	2.87	3.64	3.34
500	S4	Interval	7.39	7.75	7.69	6.54	6.91	7.62	7.14	7.75	7.72
500	S4	Exact	7.23	7.63	7.58	6.30	6.67	7.48	6.95	7.64	7.61
500	S5	Interval	5.08	5.38	5.30	7.07	5.33	5.90	5.99	7.40	7.16
500	S5	Exact	4.09	4.56	4.46	5.23	4.20	4.83	4.63	6.41	5.81
1000	S1	Interval	1.95	2.10	2.02	4.45	2.16	2.31	2.44	4.16	3.90
1000	S1	Exact	1.20	1.34	1.26	2.49	1.40	1.42	1.48	2.29	1.95
1000	S2	Interval	2.79	3.62	3.25	3.98	2.73	3.31	3.21	4.47	4.23
1000	S2	Exact	2.04	3.01	2.61	2.43	1.91	2.47	2.17	3.64	3.05
1000	S3	Interval	1.47	1.55	1.51	3.34	1.64	1.78	1.88	2.82	2.69
1000	S3	Exact	1.02	1.10	1.07	2.21	1.19	1.23	1.28	1.94	1.65
1000	S4	Interval	7.13	7.74	7.64	5.71	6.30	7.34	6.31	7.46	7.35
1000	S4	Exact	6.94	7.58	7.48	5.46	5.99	7.15	6.06	7.43	7.26
1000	S5	Interval	2.74	3.15	3.00	4.80	2.83	3.31	3.40	5.04	4.76
1000	S5	Exact	1.86	2.32	2.18	2.83	1.91	2.30	2.20	3.53	3.00
2000	S1	Interval	1.20	1.30	1.26	2.81	1.22	1.32	1.36	2.71	2.55
2000	S1	Exact	0.53	0.56	0.54	1.42	0.64	0.59	0.65	0.86	0.75
2000	S2	Interval	1.75	2.23	2.04	2.72	1.64	1.96	1.95	2.79	2.66
2000	S2	Exact	1.02	1.56	1.34	1.45	0.97	1.19	1.11	1.67	1.44
2000	S3	Interval	0.79	0.85	0.83	1.86	0.85	0.91	0.94	1.56	1.48
2000	S3	Exact	0.45	0.47	0.45	1.23	0.55	0.50	0.55	0.72	0.63
2000	S4	Interval	6.97	7.81	7.68	5.10	5.79	6.78	5.34	6.79	6.53
2000	S4	Exact	6.73	7.55	7.45	4.85	5.45	6.39	5.05	6.79	6.30
2000	S5	Interval	1.55	1.85	1.75	3.02	1.50	1.77	1.79	3.02	2.84
2000	S5	Exact	0.82	1.08	0.99	1.58	0.86	0.96	0.96	1.43	1.22

B.5 Simulation models and comparison procedures

The following specifications and complete grids accompany Section 4. Exact and Interval identify the observation scheme.

B.5.1 Event and observation models

Except in the allocation and confounding experiments, group membership is generated as $G_i \sim \text{Bernoulli}(1/2)$. Some scenarios include a long-term survivor subgroup to distinguish differences in the probability of ever experiencing the event from differences in its timing. For a subject in group g, set $T_i = \infty$ with probability c_g. With probability $1 - c_g$, the subject eventually experiences the event and its finite event time has piecewise-exponential hazard

$$\lambda_g(t) = \lambda_0 r_g(t), \qquad r_0(t) = 1. \tag{B.1}$$

The null sets $c_1 = c_0$ and $r_1(t) = 1$, leaving the observation mechanism unchanged.

The five scenarios separate the features to which the weights are intended to respond. S1 changes event timing among subjects who eventually fail, with equal cure probabilities. S2 changes cure probability while leaving the finite-event hazards equal. In S3, the treatment hazard is lower before time 2.5, whereas in S4 it is lower only after time 3. Neither S3 nor S4 has a cure fraction. S5 combines changes in event timing and long-term survival. Table B.8 gives all parameters, with probabilities d of loss to follow-up rounded to three decimal places. In S1 and S5, the reported hazard ratio concerns subjects with finite event times and need not be the population hazard ratio.

For interval assessment spacing δ and planned follow-up duration $K\delta$, the final observed assessment index is $K_i = \min\{1 + Q_i, K\}$, where $\Pr(Q_i = q) = d(1-d)^{q-1}$ for $q = 1, 2, \ldots$. Right censoring occurs at δK_i. Exact observation records T_i when $T_i \le \delta K_i$, whereas interval-censored observation records $(\delta(k-1), \delta k]$ when failure is first detected at visit k. The first such interval represents left censoring. An event-free final visit gives right censoring. Paired comparisons use the same latent event times and censoring times under the two observation maps.

Table B.8: Scenario specifications across the five-size grid. HR is the susceptible treatment-to-control hazard ratio; follow-up is $K\delta$.

Scenario	λ_0	(c_0, c_1)	Follow-up	δ	d	Treatment HR
S1	0.21	$(0.15, 0.15)$	8	1	0.14	0.65 throughout
S2	0.25	$(0.15, 0.34)$	10	1	0.12	1 throughout
S3	0.20	$(0, 0)$	6	0.5	0.034	0.65 to 2.5, then 1
S4	0.18	$(0, 0)$	8	0.5	0.030	1 to 3, then 0.55
S5	0.23	$(0.20, 0.30)$	10	1	0.035	0.82 throughout

LR, GB, PP, FH and GT denote logrank, Gehan–Breslow, Peto–Peto, Fleming–Harrington $(0, 1)$ and Gray–Tsiatis weights. LR–GT is the equal arithmetic average of the LR and GT e-processes. The primary S1 comparison uses LR and S2 uses GT. Primary GT weights use the control survival curve fixed by the simulation design, whereas the broader weight experiment uses pooled survival available before each update, truncated below at $s_{\min} = 0.12$. For each primary scenario, sample size and observation scheme, 1,000 independent null pilot datasets determine the planned information schedule. A separate set of 1,000 alternative pilot datasets selects η by maximizing mean terminal log e-value over 50 equally spaced values from -1.25 to -0.015. The selected coefficients range from -0.418 to -0.217 and remain fixed during independent evaluation. Coefficients and full calibration curves are included with the reproducibility files.

B.5.2 Procedures, information schedules and performance measures

For a selected $\mathfrak{G}_j$-measurable weight, write $U_J = \sum_{j=1}^{J} w_j(Y_j - \mu_j)$, $V_J = \sum_{j=1}^{J} w_j^2 v_j$ and $Z_J = U_J/\sqrt{V_J}$ for $V_J > 0$. Comparisons assigned to the same monitoring time are combined before a decision. The primary comparison uses four procedures:

E The e-process rejects at the first monitoring time with $E_j \geq 20$.

Fixed The terminal score test rejects once at the prespecified end of follow-up if $Z_j \leq \Phi^{-1}(0.05)$.

OBF The O'Brien–Fleming-type comparator rejects at a planned calendar assessment time τ_k, $k = 1, \ldots, L$, if $Z(\tau_k) \leq -1.96/\sqrt{t_k}$, where $t_k = \widehat{I}_k^{(0)}/\widehat{I}_L^{(0)}$ and $\widehat{I}_k^{(0)} = B_0^{-1} \sum_{b=1}^{B_0} V_b^{(0)}(\tau_k)$ is estimated from independent null pilots. The same scheduled calendar grid is used for exact and interval observations. Cumulative score and information are carried forward at visits with no new events, and zero-information looks cannot reject. This is an asymptotic comparator, not an exact level-0.05 procedure.

Repeated The terminal 5% score critical value is applied at each monitoring time, without adjustment for repeated testing.

The Fixed procedure uses the terminal version of the same accumulated table score as the e-process.

For interval observations, we additionally use the generalized logrank test of Sun (1996) as implemented by Fay and Shaw (2010). We call the R function `ictest` in the `interval` package with `scores = "logrank1"` and `method = "pclt"` for asymptotic permutation calibration. Like Fixed, it is evaluated once at the planned end of follow-up, using the same interval records and nominal level 0.05 to test for fewer treatment events than expected under the conditional null. On the common assessment grid its score equals the unweighted U_J, but its variance is the finite-population permutation variance of the pooled subject scores, rather than V_J. The permutation calibration requires exchangeability of group labels under the null, including the observation mechanism; the treatment-independent assessment and censoring rules satisfy this requirement in these simulations.

In observation columns, Interval denotes interval-censored observations and Exact denotes exact failures with right censoring. Null and power columns give rejection probability by the planned end of follow-up under the null and alternative. RC is the observed right-censored fraction under the alternative. Stopping summaries use alternative replications and calendar time. Median stop is the median first-rejection time among rejecting replications; a dash means none rejected. Mean capped stop averages the time to first rejection across all replications, using the full planned follow-up duration for each replication in which no rejection occurs. Both stopping columns for Fixed report its known planned analysis time: 8 years for S1 and 10 years for S2. Monte Carlo standard errors for rejection probabilities are $\{\widehat{p}(1-\widehat{p})/B\}^{1/2}$, where B is the number of evaluation replications. Their maximum across the reported tables is 0.0071. Paired analyses use the same latent event and censoring variables; their standard errors are the sample standard deviations of within-replication differences divided by $\sqrt{B}$.

The OBF boundary constant remains 1.96 in every reported comparison; it was not calibrated to attain 0.05 for the finite Gaussian look schedule.

B.5.3 Primary alternative-specific results

Table B.9: Primary S1 (LR) and S2 (GT) results at every total sample size.

n	Scenario	Obs.	Procedure	Null	Power	Median stop	Mean capped stop
100	S1	Interval	E	0.003	0.077	5.00	7.75
100	S1	Interval	Fixed	0.047	0.376	8.00	8.00
100	S1	Interval	OBF	0.035	0.322	4.00	6.85
100	S1	Interval	Repeated	0.125	0.537	2.00	5.10
100	S1	Exact	E	0.005	0.090	3.84	7.65
100	S1	Exact	Fixed	0.048	0.376	8.00	8.00
100	S1	Exact	OBF	0.034	0.323	4.00	6.83
100	S1	Exact	Repeated	0.277	0.659	0.75	3.63
100	S2	Interval	E	0.007	0.136	7.00	9.61
100	S2	Interval	Fixed	0.055	0.465	10.00	10.00
100	S2	Interval	OBF	0.042	0.381	7.00	8.79
100	S2	Interval	Repeated	0.163	0.634	3.00	6.00
100	S2	Exact	E	0.009	0.144	6.50	9.50
100	S2	Exact	Fixed	0.056	0.466	10.00	10.00
100	S2	Exact	OBF	0.042	0.380	7.00	8.78
100	S2	Exact	Repeated	0.335	0.731	1.20	4.25
200	S1	Interval	E	0.013	0.321	4.00	6.73
200	S1	Interval	Fixed	0.055	0.594	8.00	8.00
200	S1	Interval	OBF	0.038	0.539	4.00	5.84
200	S1	Interval	Repeated	0.125	0.738	2.00	3.86
200	S1	Exact	E	0.017	0.364	2.91	6.28
200	S1	Exact	Fixed	0.053	0.594	8.00	8.00
200	S1	Exact	OBF	0.038	0.540	4.00	5.82
200	S1	Exact	Repeated	0.314	0.837	0.56	2.26
200	S2	Interval	E	0.016	0.468	6.00	8.06
200	S2	Interval	Fixed	0.051	0.706	10.00	10.00
200	S2	Interval	OBF	0.035	0.636	6.00	7.71
200	S2	Interval	Repeated	0.168	0.825	3.00	4.55
200	S2	Exact	E	0.022	0.504	4.87	7.55
200	S2	Exact	Fixed	0.050	0.708	10.00	10.00
200	S2	Exact	OBF	0.035	0.637	6.00	7.70
200	S2	Exact	Repeated	0.372	0.892	0.92	2.64
500	S1	Interval	E	0.020	0.795	2.00	3.81
500	S1	Interval	Fixed	0.052	0.914	8.00	8.00
500	S1	Interval	OBF	0.036	0.889	3.00	3.74
500	S1	Interval	Repeated	0.132	0.958	1.00	2.04
500	S1	Exact	E	0.036	0.840	1.55	2.93
500	S1	Exact	Fixed	0.052	0.915	8.00	8.00
500	S1	Exact	OBF	0.036	0.891	3.00	3.72
500	S1	Exact	Repeated	0.380	0.980	0.29	0.83
500	S2	Interval	E	0.018	0.875	4.00	4.83
500	S2	Interval	Fixed	0.053	0.966	10.00	10.00
500	S2	Interval	OBF	0.034	0.948	5.00	5.48
500	S2	Interval	Repeated	0.163	0.985	2.00	2.60
500	S2	Exact	E	0.036	0.903	2.77	3.91
500	S2	Exact	Fixed	0.053	0.966	10.00	10.00
500	S2	Exact	OBF	0.035	0.948	5.00	5.47
500	S2	Exact	Repeated	0.437	0.993	0.51	1.18
1000	S1	Interval	E	0.015	0.968	1.00	1.97
1000	S1	Interval	Fixed	0.052	0.994	8.00	8.00
1000	S1	Interval	OBF	0.039	0.991	2.00	2.33
1000	S1	Interval	Repeated	0.127	0.998	1.00	1.27
1000	S1	Exact	E	0.046	0.980	0.80	1.19
1000	S1	Exact	Fixed	0.053	0.994	8.00	8.00
1000	S1	Exact	OBF	0.039	0.991	2.00	2.31
1000	S1	Exact	Repeated	0.401	0.999	0.16	0.34
1000	S2	Interval	E	0.020	0.987	2.00	2.76
1000	S2	Interval	Fixed	0.047	1.000	10.00	10.00

Continued on next page

Table B.9 — continued

n	Scenario	Obs.	Procedure	Null	Power	Median stop	Mean capped stop
1000	S2	Interval	OBF	0.034	1.000	4.00	3.84
1000	S2	Interval	Repeated	0.164	1.000	1.00	1.65
1000	S2	Exact	E	0.047	0.993	1.52	1.92
1000	S2	Exact	Fixed	0.047	1.000	10.00	10.00
1000	S2	Exact	OBF	0.033	1.000	4.00	3.84
1000	S2	Exact	Repeated	0.458	1.000	0.24	0.55
2000	S1	Interval	E	0.008	0.998	1.00	1.24
2000	S1	Interval	Fixed	0.050	1.000	8.00	8.00
2000	S1	Interval	OBF	0.036	1.000	2.00	1.63
2000	S1	Interval	Repeated	0.131	1.000	1.00	1.04
2000	S1	Exact	E	0.054	0.999	0.39	0.51
2000	S1	Exact	Fixed	0.050	1.000	8.00	8.00
2000	S1	Exact	OBF	0.035	1.000	2.00	1.62
2000	S1	Exact	Repeated	0.457	1.000	0.08	0.16
2000	S2	Interval	E	0.015	1.000	1.00	1.64
2000	S2	Interval	Fixed	0.055	1.000	10.00	10.00
2000	S2	Interval	OBF	0.041	1.000	3.00	2.82
2000	S2	Interval	Repeated	0.178	1.000	1.00	1.20
2000	S2	Exact	E	0.053	1.000	0.80	0.96
2000	S2	Exact	Fixed	0.055	1.000	10.00	10.00
2000	S2	Exact	OBF	0.041	1.000	3.00	2.82
2000	S2	Exact	Repeated	0.509	1.000	0.13	0.29

B.5.4 Interval-censored benchmark

The following comparison covers every S1–S5 and sample-size combination. The common-grid calculation was checked against the `interval` implementation before evaluation. Table B.10 gives the $n = 500$ results, with the full grid in Table B.5. Null rejection for the Fay–Shaw benchmark (Sun, 1996; Fay and Shaw, 2010) ranges from 0.0448 to 0.0548, compared with 0.0446 to 0.0554 for Fixed LR. The scores agree throughout evaluation, but the different variance estimates produce 250 rejection disagreements among 250,000 interval-record datasets (0.10%).

Table B.10: Fixed-analysis interval-censored benchmark at $n = 500$. Δ is Fay–Shaw power minus Fixed LR power, with its paired Monte Carlo SE.

Scenario	Fixed LR		Fay–Shaw		Δ	Paired SE
	Null	Power	Null	Power		
S1	0.0462	0.9030	0.0462	0.9026	-0.0004	0.0003
S2	0.0476	0.9542	0.0476	0.9540	-0.0002	0.0002
S3	0.0512	0.6518	0.0512	0.6516	-0.0002	0.0004
S4	0.0478	0.5956	0.0480	0.5954	-0.0002	0.0003
S5	0.0492	0.9234	0.0492	0.9228	-0.0006	0.0003

B.5.5 Alternative-specific weight comparisons

Both weight-comparison experiments use raw LR, PP, FH and GT weights 1, $\widehat{S}_{j-1}$, $1-\widehat{S}_{j-1}$ and $1/\max(\widehat{S}_{j-1}, 0.12)$, respectively. For each dataset and observation scheme, let N_{first} be the total eligible size in its first comparison containing at least one observed event. The raw GB weight is

$$\widetilde{w}_j^{\mathrm{GB}} = \frac{N_j}{N_{\text{first}}}.$$

Zero-event windows are omitted from the stored comparisons; the first stored comparison need not have positive conditional variance. The denominator is determined separately for each dataset and scheme and then retained throughout that dataset. It need not equal total enrollment and is not fixed at 500. If there is no observed event, the process remains one. These raw weights are implemented by `all_weight_matrices` in `simulations/code/run_practical_battery.R`.

The weight comparison in Table B.11 uses 1,000 independent null pilot datasets to normalize the raw weights and 1,000 independent alternative pilot datasets to choose their coefficients in every S1–S5 sample-size and scheme combination. Let J_b be the number of updates through the planned end of follow-up in pilot dataset b. For raw weight $\widetilde{w}$, define $I_b(\widetilde{w}) = \sum_{j=1}^{J_b} \widetilde{w}_{bj}^2 v_{bj}$. The normalized weight is $w = a_{\widetilde{w}} \widetilde{w}$, where

$$a_{\widetilde{w}} = \left\{ \frac{\sum_{b=1}^{1000} I_b(1)}{\sum_{b=1}^{1000} I_b(\widetilde{w})} \right\}^{1/2}.$$

The coefficient grid is searched on the independent alternative pilots. Scale constants and selected coefficients remain fixed during independent evaluation. In the fixed-specification experiment, the same normalization formula is applied only to the S1 null pilots at $n = 500$, separately by observation scheme. Those multipliers are then retained across sample sizes and alternatives, while each dataset supplies its own N_{first}. Table B.11 shows how the timing of the alternative changes the relative performance of the weights. At $n = 500$ under interval censoring, PP has power 0.668 in the early-effect scenario S3 compared with 0.595 for LR. In the delayed-effect scenario S4, FH has power 0.681 compared with 0.257 for LR. GT has power 0.874 in S2 compared with 0.803 for LR. Thus the difference between weights can be substantial even when the observation records are the same. The maximum estimated null rejection across the weighted processes is 0.052.

Table B.11: Weighted e-process power across the common sample-size grid. Max null is the largest null rejection among the six weights.

n	Obs.	Scenario	LR	GB	PP	FH	GT	LR–GT	Max null
100	Interval	S1	0.059	0.048	0.059	0.002	0.039	0.049	0.003
100	Interval	S2	0.112	0.027	0.046	0.088	0.141	0.128	0.009
100	Interval	S3	0.002	0.015	0.016	0.000	0.000	0.000	0.001
100	Interval	S4	0.000	0.000	0.000	0.033	0.013	0.004	0.003
100	Interval	S5	0.068	0.045	0.049	0.015	0.045	0.056	0.004
100	Exact	S1	0.068	0.055	0.069	0.011	0.045	0.055	0.004
100	Exact	S2	0.104	0.031	0.054	0.115	0.138	0.122	0.010
100	Exact	S3	0.003	0.025	0.014	0.000	0.000	0.000	0.002
100	Exact	S4	0.000	0.000	0.000	0.028	0.011	0.003	0.003
100	Exact	S5	0.073	0.051	0.054	0.028	0.047	0.062	0.006
200	Interval	S1	0.317	0.265	0.294	0.071	0.262	0.292	0.011
200	Interval	S2	0.371	0.192	0.257	0.370	0.465	0.428	0.023
200	Interval	S3	0.062	0.169	0.170	0.000	0.004	0.029	0.007
200	Interval	S4	0.022	0.000	0.000	0.204	0.132	0.080	0.010
200	Interval	S5	0.331	0.248	0.263	0.141	0.318	0.329	0.012
200	Exact	S1	0.359	0.303	0.339	0.151	0.312	0.341	0.018
200	Exact	S2	0.397	0.206	0.272	0.449	0.480	0.452	0.030
200	Exact	S3	0.096	0.221	0.183	0.000	0.003	0.040	0.010
200	Exact	S4	0.028	0.000	0.000	0.201	0.153	0.097	0.013
200	Exact	S5	0.363	0.271	0.288	0.236	0.357	0.367	0.019
500	Interval	S1	0.787	0.698	0.742	0.470	0.793	0.792	0.021
500	Interval	S2	0.803	0.591	0.679	0.821	0.874	0.856	0.018
500	Interval	S3	0.595	0.660	0.668	0.000	0.227	0.492	0.021

Continued on next page

Table B.11 — continued

n	Obs.	Scenario	LR	GB	PP	FH	GT	LR–GT	Max null
500	Interval	S4	0.257	0.005	0.033	0.681	0.563	0.479	0.024
500	Interval	S5	0.781	0.685	0.707	0.614	0.804	0.800	0.025
500	Exact	S1	0.837	0.739	0.787	0.723	0.852	0.849	0.039
500	Exact	S2	0.814	0.606	0.690	0.887	0.891	0.870	0.040
500	Exact	S3	0.675	0.720	0.721	0.007	0.208	0.571	0.036
500	Exact	S4	0.277	0.007	0.041	0.683	0.588	0.508	0.039
500	Exact	S5	0.825	0.719	0.744	0.783	0.854	0.845	0.043
1000	Interval	S1	0.960	0.919	0.941	0.856	0.975	0.970	0.019
1000	Interval	S2	0.966	0.873	0.919	0.970	0.983	0.979	0.017
1000	Interval	S3	0.940	0.927	0.933	0.012	0.832	0.921	0.023
1000	Interval	S4	0.576	0.073	0.176	0.919	0.850	0.803	0.026
1000	Interval	S5	0.958	0.917	0.927	0.921	0.976	0.970	0.023
1000	Exact	S1	0.975	0.933	0.954	0.972	0.988	0.984	0.048
1000	Exact	S2	0.967	0.877	0.918	0.987	0.987	0.984	0.049
1000	Exact	S3	0.963	0.943	0.952	0.154	0.839	0.944	0.040
1000	Exact	S4	0.582	0.083	0.190	0.908	0.861	0.816	0.041
1000	Exact	S5	0.969	0.928	0.938	0.980	0.985	0.982	0.051
2000	Interval	S1	0.997	0.993	0.996	0.989	1.000	0.999	0.014
2000	Interval	S2	0.997	0.982	0.992	0.999	0.999	0.999	0.012
2000	Interval	S3	1.000	0.996	0.997	0.230	0.999	1.000	0.025
2000	Interval	S4	0.854	0.284	0.455	0.991	0.976	0.969	0.020
2000	Interval	S5	0.998	0.993	0.996	0.997	1.000	1.000	0.018
2000	Exact	S1	0.998	0.995	0.997	1.000	1.000	1.000	0.047
2000	Exact	S2	0.998	0.982	0.992	1.000	1.000	1.000	0.052
2000	Exact	S3	1.000	0.998	0.999	0.832	1.000	1.000	0.047
2000	Exact	S4	0.847	0.295	0.448	0.991	0.978	0.972	0.042
2000	Exact	S5	0.999	0.995	0.997	1.000	1.000	1.000	0.048

B.5.6 Fixed-specification stopping times

The $n = 500$ subset below accompanies the main-text power comparison; the complete five-size power and duration grids are in Tables B.6 and B.7.

Table B.12: Mean capped stopping time in calendar years for fixed practical designs and prequential comparators at total $n = 500$.

Scenario	Obs.	LR	FH	Cmix	Wmix	PQ	PQ^{-}
S1	Interval	4.00	6.52	4.49	4.68	6.04	5.83
S1	Exact	3.08	4.81	3.50	3.56	4.67	4.27
S2	Interval	5.00	6.08	5.88	5.63	7.08	6.82
S2	Exact	4.18	4.47	5.02	4.48	6.63	6.00
S3	Interval	3.02	5.18	3.36	3.54	4.36	4.24
S3	Exact	2.43	4.27	2.75	2.87	3.64	3.34
S4	Interval	7.39	6.54	7.62	7.14	7.75	7.72
S4	Exact	7.23	6.30	7.48	6.95	7.64	7.61
S5	Interval	5.08	7.07	5.90	5.99	7.40	7.16
S5	Exact	4.09	5.23	4.83	4.63	6.41	5.81

B.5.7 Sample size, censoring and treatment allocation

We examine how sample size, loss to follow-up and treatment allocation affect performance while holding the event-time model fixed. Each of the five sample sizes is combined with two treatment-assignment probabilities, $p_T \in \{1/2, 1/3\}$, and two probabilities of loss to follow-up, $d \in \{0.08, 0.20\}$. Analyzing each combination under exact and annual interval observation gives $5 \times 2 \times 2 \times 2 = 40$ cases. The balanced allocation, $p_T = 1/2$, gives equal probabilities of assignment to treatment and control; $p_T = 1/3$ assigns one subject to treatment for every two controls on average. Realized group sizes remain random. Both groups have cure probability 0.15, the susceptible control hazard is 0.21, and the susceptible treatment-to-control hazard ratio is 0.70 under the alternative. Follow-up ends at year 8.

Loss to follow-up follows the shifted geometric mechanism in Section B.5.1. The earliest censoring time is year 2, with probability d, and each subsequent year has the same conditional probability of censoring, subject to the year-8 limit. We use logrank weights and retain one reference coefficient for each observation scheme throughout the 40 cases. These coefficients are selected to maximize pilot mean terminal log evidence at $n = 500$, $p_T = 1/2$, $d = 0.14$ and hazard ratio 0.70, the middle of the three hazard ratios 0.80, 0.70 and 0.60 considered in Section B.5.8. For each case, 1,000 independent null pilot datasets determine the information schedule for the O'Brien–Fleming comparator. Thus the e-process coefficient stays fixed as the study design changes.

The factorial e-process null rejection ranges from 0.000 to 0.052, compared with 0.115 to 0.447 for unadjusted repeated testing (Tables B.13 and B.14).

Table B.13: Sample size, allocation and loss to follow-up under interval censoring.

n	p_T	d	RC	E null	E power	Fixed power	OBF power	Repeated null
100	0.33	0.08	0.424	0.001	0.019	0.263	0.211	0.130
100	0.33	0.20	0.519	0.000	0.008	0.250	0.194	0.115
100	0.50	0.08	0.444	0.002	0.031	0.308	0.261	0.128
100	0.50	0.20	0.534	0.001	0.014	0.296	0.242	0.117
200	0.33	0.08	0.426	0.008	0.158	0.436	0.382	0.127
200	0.33	0.20	0.519	0.004	0.108	0.396	0.337	0.117
200	0.50	0.08	0.443	0.007	0.198	0.479	0.426	0.125
200	0.50	0.20	0.535	0.006	0.145	0.451	0.387	0.126
500	0.33	0.08	0.425	0.020	0.598	0.761	0.720	0.128
500	0.33	0.20	0.518	0.015	0.526	0.726	0.670	0.124
500	0.50	0.08	0.443	0.021	0.654	0.819	0.782	0.131
500	0.50	0.20	0.535	0.019	0.589	0.772	0.731	0.116
1000	0.33	0.08	0.425	0.020	0.882	0.964	0.949	0.130
1000	0.33	0.20	0.518	0.021	0.833	0.942	0.921	0.115
1000	0.50	0.08	0.443	0.016	0.903	0.975	0.963	0.127
1000	0.50	0.20	0.535	0.017	0.883	0.963	0.952	0.121
2000	0.33	0.08	0.425	0.015	0.986	1.000	0.999	0.139
2000	0.33	0.20	0.518	0.011	0.980	0.998	0.998	0.123
2000	0.50	0.08	0.443	0.012	0.989	1.000	0.999	0.133
2000	0.50	0.20	0.535	0.011	0.988	0.999	0.999	0.129

Table B.14: Sample size, allocation and loss to follow-up under exact event observation with right censoring.

n	p_T	d	RC	E null	E power	Fixed power	OBF power	Repeated null
100	0.33	0.08	0.424	0.001	0.021	0.262	0.207	0.243
100	0.33	0.20	0.519	0.000	0.009	0.249	0.191	0.222
100	0.50	0.08	0.444	0.003	0.035	0.312	0.263	0.284
100	0.50	0.20	0.534	0.001	0.016	0.297	0.242	0.268
200	0.33	0.08	0.426	0.012	0.184	0.436	0.380	0.290
200	0.33	0.20	0.519	0.006	0.127	0.395	0.333	0.267
200	0.50	0.08	0.443	0.012	0.231	0.482	0.428	0.309

Continued on next page

Table B.14 — continued

n	p_T	d	RC	E null	E power	Fixed power	OBF power	Repeated null
200	0.50	0.20	0.535	0.008	0.166	0.449	0.388	0.317
500	0.33	0.08	0.425	0.029	0.662	0.763	0.721	0.324
500	0.33	0.20	0.518	0.029	0.584	0.724	0.672	0.333
500	0.50	0.08	0.443	0.035	0.718	0.823	0.783	0.377
500	0.50	0.20	0.535	0.036	0.648	0.777	0.732	0.369
1000	0.33	0.08	0.425	0.046	0.919	0.964	0.950	0.370
1000	0.33	0.20	0.518	0.040	0.877	0.943	0.921	0.362
1000	0.50	0.08	0.443	0.044	0.935	0.975	0.963	0.410
1000	0.50	0.20	0.535	0.041	0.920	0.964	0.952	0.400
2000	0.33	0.08	0.425	0.052	0.993	1.000	0.999	0.405
2000	0.33	0.20	0.518	0.050	0.990	0.998	0.998	0.406
2000	0.50	0.08	0.443	0.048	0.995	1.000	1.000	0.446
2000	0.50	0.20	0.535	0.048	0.994	0.999	0.999	0.447

B.5.8 Interval assessment frequency and effect-size sensitivity

To study assessment frequency, we keep the event-time model and calendar follow-up fixed and change how often events can be detected. At every sample size, both groups have cure probability 0.15, the susceptible control hazard is 0.21 per year, and the alternative hazard ratio is 0.70. Treatment assignment has probability 1/2. Follow-up lasts 8 years, with assessments spaced at $\delta \in \{0.5, 1, 2\}$ years, giving $K = 16$, 8 or 4 scheduled assessments. Independent exponential censoring has rate 0.085 and is administratively truncated at year 8. Only assessments completed before censoring contribute an observed interval, so longer gaps between visits can leave a longer final period without an assessment.

We use logrank weights and retain the interval-observation coefficient from the annual, $n = 500$ reference design in Section B.5.7. The coefficient is not retuned when the spacing or sample size changes. Each case uses 1,000 independent null pilot datasets for its information schedule. In Table B.15, median stop refers to the e-process.

Table B.15: Interval assessment frequency at every total sample size.

n	δ	K	RC	E null	E power	Fixed power	OBF power	Repeated null	Median stop
100	0.5	16	0.510	0.002	0.020	0.283	0.241	0.165	5.00
100	1.0	8	0.520	0.001	0.013	0.274	0.226	0.130	6.00
100	2.0	4	0.542	0.000	0.008	0.271	0.206	0.096	6.00
200	0.5	16	0.510	0.011	0.182	0.453	0.417	0.162	4.00
200	1.0	8	0.521	0.006	0.145	0.442	0.383	0.136	5.00
200	2.0	4	0.542	0.004	0.102	0.411	0.337	0.094	6.00
500	0.5	16	0.510	0.023	0.615	0.771	0.739	0.154	2.50
500	1.0	8	0.520	0.017	0.597	0.782	0.737	0.127	3.00
500	2.0	4	0.543	0.014	0.519	0.747	0.683	0.097	4.00
1000	0.5	16	0.509	0.022	0.889	0.966	0.959	0.161	1.50
1000	1.0	8	0.521	0.020	0.871	0.957	0.943	0.137	2.00
1000	2.0	4	0.542	0.015	0.839	0.944	0.921	0.096	2.00
2000	0.5	16	0.510	0.015	0.989	0.999	0.999	0.163	1.00
2000	1.0	8	0.520	0.009	0.984	1.000	0.999	0.128	1.00
2000	2.0	4	0.543	0.007	0.981	0.999	0.997	0.099	2.00

The effect-size experiment examines the cost of choosing a coefficient for a reference effect that differs from the true effect. The latency family changes the timing of finite events while keeping cure probability at 0.15 in both groups. Its susceptible control hazard is 0.21, follow-up is 8 years, and the treatment hazard ratio is 0.80, 0.70 or 0.60. The cure family instead keeps the susceptible hazard at 0.25 in both groups and changes the treatment cure probability to 0.25, 0.34 or 0.43, compared with 0.15 in the control group.

Its follow-up is 10 years. The respective shifted geometric censoring probabilities are 0.14 and 0.12. Treatment assignment has probability 1/2, and we consider exact and annual interval observations at every sample size.

We use logrank weights for the latency family and Gray–Tsiatis weights for the cure family. A reference coefficient is selected to maximize pilot mean terminal log evidence separately for each family and observation scheme at $n = 500$, using hazard ratio 0.70 or treatment cure probability 0.34. These are the middle values in the two effect grids. The reference coefficients are then kept fixed across sample sizes and true effects. Table B.2 compares their performance with coefficients selected by the same objective of maximizing mean terminal log evidence using 1,000 independent alternative pilot datasets at the true effect and sample size. Both choices are evaluated on the same datasets. Mean terminal $\log E$ averages the final log e-value over all alternative replications.

Table B.16: Effect sensitivity using the coefficient fixed at the $n = 500$ middle design alternative. S1 and S2 denote the latency and cure families, with susceptible HR and treatment cure probability as the respective effects. Their effects vary from the primary specifications in Table B.8.

n	Family	Effect	Obs.	η	Power	Median stop	Capped stop	Mean terminal $\log E$
100	S1	0.60	Interval	-0.343	0.050	5.00	7.86	1.085
100	S1	0.60	Exact	-0.317	0.052	4.43	7.82	1.082
100	S1	0.70	Interval	-0.343	0.022	5.00	7.94	0.610
100	S1	0.70	Exact	-0.317	0.025	4.45	7.91	0.612
100	S1	0.80	Interval	-0.343	0.009	5.00	7.98	0.157
100	S1	0.80	Exact	-0.317	0.011	4.19	7.96	0.161
100	S2	0.25	Interval	-0.217	0.041	7.00	9.89	0.060
100	S2	0.25	Exact	-0.217	0.045	6.59	9.85	0.068
100	S2	0.34	Interval	-0.217	0.146	7.00	9.59	1.179
100	S2	0.34	Exact	-0.217	0.156	6.57	9.46	1.176
100	S2	0.43	Interval	-0.217	0.332	7.00	8.95	2.183
100	S2	0.43	Exact	-0.217	0.331	6.14	8.77	2.156
200	S1	0.60	Interval	-0.343	0.380	4.00	6.58	2.253
200	S1	0.60	Exact	-0.317	0.416	3.32	6.17	2.243
200	S1	0.70	Interval	-0.343	0.171	4.00	7.38	1.175
200	S1	0.70	Exact	-0.317	0.194	3.37	7.16	1.182
200	S1	0.80	Interval	-0.343	0.075	4.00	7.73	0.295
200	S1	0.80	Exact	-0.317	0.095	3.42	7.59	0.303
200	S2	0.25	Interval	-0.217	0.149	6.00	9.43	0.174
200	S2	0.25	Exact	-0.217	0.173	5.35	9.22	0.194
200	S2	0.34	Interval	-0.217	0.469	6.00	8.06	2.358
200	S2	0.34	Exact	-0.217	0.499	4.83	7.57	2.356
200	S2	0.43	Interval	-0.217	0.826	5.00	5.97	4.457
200	S2	0.43	Exact	-0.217	0.840	4.05	5.33	4.403
500	S1	0.60	Interval	-0.343	0.900	2.00	3.12	5.580
500	S1	0.60	Exact	-0.317	0.922	1.53	2.36	5.566
500	S1	0.70	Interval	-0.343	0.621	3.00	5.03	3.065
500	S1	0.70	Exact	-0.317	0.685	2.02	4.21	3.076
500	S1	0.80	Interval	-0.343	0.287	3.00	6.67	0.742
500	S1	0.80	Exact	-0.317	0.349	2.12	6.10	0.763
500	S2	0.25	Interval	-0.217	0.355	5.00	8.25	0.392
500	S2	0.25	Exact	-0.217	0.429	3.67	7.47	0.445
500	S2	0.34	Interval	-0.217	0.890	4.00	4.69	5.996
500	S2	0.34	Exact	-0.217	0.914	2.69	3.78	5.992
500	S2	0.43	Interval	-0.217	0.996	2.00	2.72	11.136
500	S2	0.43	Exact	-0.217	0.998	1.78	2.05	11.011
1000	S1	0.60	Interval	-0.343	0.997	1.00	1.51	11.249
1000	S1	0.60	Exact	-0.317	0.999	0.74	0.90	11.222
1000	S1	0.70	Interval	-0.343	0.895	2.00	2.78	6.154
1000	S1	0.70	Exact	-0.317	0.930	1.10	1.89	6.186
1000	S1	0.80	Interval	-0.343	0.494	2.00	5.44	1.429

Continued on next page

Table B.16 — continued

n	Family	Effect	Obs.	η	Power	Median stop	Capped stop	Mean terminal $\log E$
1000	S1	0.80	Exact	-0.317	0.593	1.45	4.39	1.470
1000	S2	0.25	Interval	-0.217	0.519	4.00	6.99	0.830
1000	S2	0.25	Exact	-0.217	0.616	2.58	5.80	0.938
1000	S2	0.34	Interval	-0.217	0.987	2.00	2.72	12.172
1000	S2	0.34	Exact	-0.217	0.991	1.49	1.88	12.166
1000	S2	0.43	Interval	-0.217	1.000	2.00	1.59	22.304
1000	S2	0.43	Exact	-0.217	1.000	0.92	1.03	22.057
2000	S1	0.60	Interval	-0.343	1.000	1.00	1.06	22.519
2000	S1	0.60	Exact	-0.317	1.000	0.36	0.42	22.460
2000	S1	0.70	Interval	-0.343	0.991	1.00	1.50	12.440
2000	S1	0.70	Exact	-0.317	0.995	0.55	0.76	12.501
2000	S1	0.80	Interval	-0.343	0.687	2.00	4.00	2.861
2000	S1	0.80	Exact	-0.317	0.796	0.85	2.66	2.949
2000	S2	0.25	Interval	-0.217	0.658	3.00	5.72	1.737
2000	S2	0.25	Exact	-0.217	0.765	1.67	4.17	1.955
2000	S2	0.34	Interval	-0.217	1.000	1.00	1.66	23.894
2000	S2	0.34	Exact	-0.217	1.000	0.80	0.98	23.886
2000	S2	0.43	Interval	-0.217	1.000	1.00	1.09	44.966
2000	S2	0.43	Exact	-0.217	1.000	0.48	0.54	44.467

B.5.9 Covariate stratification, delayed entry and confounding

This experiment examines whether stratification can account for a prognostic covariate that also affects treatment assignment when subjects enter observation at different times. We use every sample size and eight annual intervals. The binary covariate $Z_i \sim \text{Bernoulli}(1/2)$ multiplies the susceptible baseline hazard 0.17 by 2.6 when $Z_i = 1$. Treatment probabilities are 0.28 for $Z_i = 1$ and 0.72 for $Z_i = 0$. Consequently, the treatment group contains more subjects from the lower-risk stratum. Pooling the strata can therefore suggest that treatment lowers the event rate even when treatment has no effect within either stratum.

Subjects enter at $A_i = 0$, 1 or 2 with probabilities 0.55, 0.30 and 0.15 and must be event-free at entry. Their event times are generated from an origin-time cure model conditioned on survival to entry. The origin-population cure probability is $c = 0.12$, but among subjects entering at $A_i = a$ it is $c/\{c + (1-c)e^{-\lambda_{g,z}a}\}$. A susceptible subject has $T_i = a + \text{Exp}(\lambda_{g,z})$, where $\lambda_{g,z} = 0.17(2.6)^z(0.65)^g$ under the alternative and the treatment multiplier is 1 under the null. This conditioning accounts for the increasing proportion of cured subjects among those surviving to a later entry time. Independent censoring occurs after an exponential time with rate 0.05 following entry. A subject contributes to an annual comparison only after entry and if the assessment interval is completed before censoring.

At a given calendar interval, subjects in the same Z stratum have the same null event probability regardless of their entry cohort. We analyze each dataset twice using logrank weights, first within the two Z strata and then after pooling their counts. For each sample size, 1,000 null and 1,000 alternative pilot datasets determine the analysis-specific information schedules and coefficients. Pooling deliberately removes the adjustment needed for the conditional equality assumption. Stratified e-process null rejection ranges from 0.001 to 0.021, compared with 0.179–0.842 after pooling (Table B.17). Controlling repeated monitoring therefore does not correct the confounding in the pooled comparison.

Table B.17: Covariate and delayed-entry experiment.

n	Analysis	E null	E power	Fixed null	Fixed power	OBF null	OBF power	Repeated null
100	Stratified	0.001	0.022	0.049	0.295	0.035	0.259	0.146
100	Pooled	0.179	0.601	0.311	0.758	0.280	0.732	0.523
200	Stratified	0.007	0.182	0.047	0.452	0.034	0.415	0.138
200	Pooled	0.343	0.890	0.496	0.952	0.465	0.948	0.714
500	Stratified	0.019	0.647	0.054	0.783	0.036	0.762	0.150
500	Pooled	0.584	0.998	0.835	1.000	0.831	1.000	0.946
1000	Stratified	0.021	0.910	0.048	0.966	0.039	0.960	0.144
1000	Pooled	0.719	1.000	0.981	1.000	0.984	1.000	0.996
2000	Stratified	0.017	0.993	0.046	1.000	0.035	1.000	0.139
2000	Pooled	0.842	1.000	1.000	1.000	1.000	1.000	1.000

C Clinical evidence summaries and observation diagnostics

C.1 Complete evidence summaries

Table C.1 reports evidence summaries for every weight and width. The reference widths are $q = 0.25$ years for CAV and $q = 6$ months for PsA. All processes continue after crossing the reference threshold. The analyses reconstruct aligned follow-up retrospectively and rely on the working observation assumptions described in Section 5. Counts of informative comparisons and events are in Table 4; retaining every first detection does not make every table informative.

Table C.1: Complete descriptive clinical evidence summaries. CAV times and widths q are in years; PsA times and widths are in months. First crossing is the first time with $E \geq 20$, with a dash for no crossing. These crossings do not establish that the exact guarantee applies to the cohorts' irregular observation mechanisms.

Application	q	Weight	max E	Terminal E	First crossing
CAV, $q = 0.10$: $(n_0, n_1) = (270, 313)$; events = 214; comparisons/times = 521/95					
CAV	0.10	LR	135.80	12.26	4.25 years
CAV	0.10	GB	3.18	2.87	–
CAV	0.10	PP	2.06	2.00	–
CAV	0.10	FH	51.62	5.19	5.05 years
CAV	0.10	GT	1.07	1.03×10^{-17}	–
CAV	0.10	LR–GT	67.90	6.13	5.05 years
CAV, $q = 0.25$: $(n_0, n_1) = (270, 313)$; events = 214; comparisons/times = 329/51					
CAV	0.25	LR	369.11	34.69	4.38 years
CAV	0.25	GB	13.49	11.42	–
CAV	0.25	PP	17.13	14.93	–
CAV	0.25	FH	34.97	3.45	5.12 years
CAV	0.25	GT	1.01	5.14×10^{-15}	–
CAV	0.25	LR–GT	184.56	17.35	5.12 years
CAV, $q = 0.50$: $(n_0, n_1) = (270, 313)$; events = 214; comparisons/times = 257/31					
CAV	0.50	LR	133.75	20.57	5.25 years
CAV	0.50	GB	33.48	28.56	5.25 years
CAV	0.50	PP	23.77	19.51	7.25 years

Continued on next page

Table C.1 continued

Application	q	Weight	max E	Terminal E	First crossing
CAV	0.50	FH	11.61	1.61	–
CAV	0.50	GT	1.00	1.25×10^{-17}	–
CAV	0.50	LR–GT	66.88	10.29	5.25 years
PsA, $q = 3$: $(n_0, n_1) = (120, 65)$; events = 72; comparisons/times = 16/7					
PsA	3	LR	2.08	2.08	–
PsA	3	GB	1.34	1.34	–
PsA	3	PP	1.41	1.41	–
PsA	3	FH	1.71	1.71	–
PsA	3	GT	3.84	3.84	–
PsA	3	LR–GT	2.96	2.96	–
PsA, $q = 6$: $(n_0, n_1) = (120, 65)$; events = 72; comparisons/times = 8/4					
PsA	6	LR	11.22	11.22	–
PsA	6	GB	3.55	3.55	–
PsA	6	PP	6.73	6.73	–
PsA	6	FH	1.71	1.71	–
PsA	6	GT	23.63	23.63	18 months
PsA	6	LR–GT	17.43	17.43	–
PsA, $q = 12$: $(n_0, n_1) = (120, 65)$; events = 72; comparisons/times = 4/2					
PsA	12	LR	7.32	7.32	–
PsA	12	GB	5.00	5.00	–
PsA	12	PP	6.42	6.42	–
PsA	12	FH	1.15	1.15	–
PsA	12	GT	8.78	8.78	–
PsA	12	LR–GT	8.05	8.05	–

C.2 Observed alternative magnitudes

Table C.2 gives the extrema of ηw_j over the actual tables and coefficient components. The intervals describe extrema, not a continuum of fitted coefficients. We also restrict the calculation to informative tables, since a large weight in a zero-variance table has no effect on evidence. The LR–GT process averages the complete LR and GT processes and has no single table weight or coefficient product.

For reference CAV, the GT cap is attained in informative tables; ηw_j reaches ± 5.776, whereas LR reaches only ± 0.693. For reference PsA, the largest GT weight over all tables is 4.071, attained in an uninformative table. The largest informative-table weight is 2.326, giving a smallest group-1 event-odds alternative of $\exp(-1.613) \simeq 0.199$, compared with 0.5 for LR. These are comparisons of complete working alternatives, without a common normalization across weights.

Table C.2: Observed ranges of ηw_j, rounded to three decimals. "Informative" restricts to positive hypergeometric conditional variance. Widths are years for CAV and months for PsA. Zero weights are retained.

Application	q	Weight	All comparisons	Informative comparisons
CAV	0.10	LR	$[-0.693, 0.693]$	$[-0.693, 0.693]$

Continued on next page

Table C.2 continued

Application	q	Weight	All comparisons	Informative comparisons
CAV	0.10	GB	$[-0.533, 0.533]$	$[-0.533, 0.533]$
CAV	0.10	PP	$[-0.693, 0.693]$	$[-0.693, 0.693]$
CAV	0.10	FH	$[-0.693, 0.693]$	$[-0.693, 0.693]$
CAV	0.10	GT	$[-5.776, 5.776]$	$[-5.776, 5.776]$
CAV	0.25	LR	$[-0.693, 0.693]$	$[-0.693, 0.693]$
CAV	0.25	GB	$[-0.533, 0.533]$	$[-0.533, 0.533]$
CAV	0.25	PP	$[-0.693, 0.693]$	$[-0.693, 0.693]$
CAV	0.25	FH	$[-0.693, 0.693]$	$[-0.693, 0.693]$
CAV	0.25	GT	$[-5.776, 5.776]$	$[-5.776, 5.776]$
CAV	0.50	LR	$[-0.693, 0.693]$	$[-0.693, 0.693]$
CAV	0.50	GB	$[-0.563, 0.563]$	$[-0.563, 0.563]$
CAV	0.50	PP	$[-0.693, 0.693]$	$[-0.693, 0.693]$
CAV	0.50	FH	$[-0.693, 0.693]$	$[-0.693, 0.693]$
CAV	0.50	GT	$[-5.776, 5.776]$	$[-5.776, 5.776]$
PsA	3	LR	$[-0.693, -0.223]$	$[-0.693, -0.223]$
PsA	3	GB	$[-0.485, -0.001]$	$[-0.485, -0.007]$
PsA	3	PP	$[-0.693, 0.000]$	$[-0.693, -0.050]$
PsA	3	FH	$[-0.693, 0.000]$	$[-0.539, 0.000]$
PsA	3	GT	$[-5.776, -0.223]$	$[-3.108, -0.223]$
PsA	6	LR	$[-0.693, -0.223]$	$[-0.693, -0.223]$
PsA	6	GB	$[-0.624, -0.001]$	$[-0.624, -0.008]$
PsA	6	PP	$[-0.693, -0.055]$	$[-0.693, -0.096]$
PsA	6	FH	$[-0.523, 0.000]$	$[-0.395, 0.000]$
PsA	6	GT	$[-2.822, -0.223]$	$[-1.613, -0.223]$
PsA	12	LR	$[-0.693, -0.223]$	$[-0.693, -0.223]$
PsA	12	GB	$[-0.693, -0.001]$	$[-0.693, -0.009]$
PsA	12	PP	$[-0.693, -0.138]$	$[-0.693, -0.138]$
PsA	12	FH	$[-0.263, 0.000]$	$[-0.263, 0.000]$
PsA	12	GT	$[-1.118, -0.223]$	$[-1.118, -0.223]$

The accompanying diagnostics recompute all 1,140 saved weight-specific path values directly from the fixed-margin cells using hypergeometric probabilities. The largest absolute difference in log evidence is 1.42×10^{-13}. This checks the evidence calculation from the saved tables; it does not independently validate the original patient-level preprocessing. The files also give every conditional variance, full-precision coefficient range and the component log increments underlying Table 5.

C.3 Synthetic irregular-visit stress tests

We use the unmodified `cav_analysis` and `psor_analysis` functions from the clinical program, including endpoint rounding, last-visit selection, stopping at the first retained event, grouping by nominal endpoints, batching, pooled survival updates and coefficient mixtures. The functions are loaded without running the historical application driver. Synthetic data have one homogeneous prognostic stratum, independent exponential event times with the same rate in both groups, visits independent of event times, and no loss to follow-up. Thus the latent event-time distributions agree between groups in every mechanism. The simulation does not include informative death, attendance depending on disease activity, residual prognostic heterogeneity or staggered recruitment.

For CAV, there are 270 group-0 and 313 group-1 subjects, event rate 0.12 per year, six nominal annual visits and rounding width $q = 0.25$ years. The common-grid mechanism uses exact annual visits. Independent jitter adds independent Uniform$(-0.12, 0.12)$ offsets. Differential jitter adds Uniform$(-0.12, 0)$ offsets in group 0 and Uniform$(0, 0.12)$ offsets in group 1. All offsets remain within the same rounding bin, so grouping conceals differences between the actual intervals.

For PsA, there are 120 group-0 and 65 group-1 subjects, event rate 0.025 per month and four six-month windows. Each window contains two visits; the actual algorithm retains the last one. Under the common grid, the visits occur five and six months after the window begins. Independent jitter gives both visits independent Uniform$(0.2, 5.8)$ offsets. Differential jitter uses Uniform$(3.2, 5.8)$ offsets in group 0 and Uniform$(0.2, 2.8)$ offsets in group 1. The nominal bins coincide, but group-1 patients are examined earlier. The direction of the PsA mixture is fixed before simulation.

Each application and mechanism has 2,000 independent replications. Within a replication, latent event times and the underlying uniform draws are shared across mechanisms. We examine the original LR and GT coefficient mixtures without fitting or selecting a weight. Under the exponential model, an actual interval $(l, r]$ has conditional event probability $1 - \exp\{-\lambda(r - l)\}$. The saved diagnostics record its range within each constructed comparison. A common grid has range zero. Unequal realized probabilities conditional on the actual visit times do not by themselves exclude exchangeability after averaging over group-independent schedules; the relevant conditioning must be specified. Differential schedules can associate interval length with group despite equal latent hazards.

Table C.3: Null stress tests using the actual clinical grouping functions. Each crossing frequency uses 2,000 independent replications; brackets give 95% Wilson Monte Carlo intervals. The final column is the median, over replications, of the largest within-table range of actual interval event probabilities. These are synthetic mechanism checks, not estimates of clinical false-positive rates.

Application	Visit mechanism	LR	GT	Probability range
CAV	Common grid	0.0165 $[0.0118, 0.0231]$	0.0155 $[0.0109, 0.0219]$	0
CAV	Independent jitter	0.0170 $[0.0122, 0.0237]$	0.0150 $[0.0105, 0.0213]$	0.0493
CAV	Differential jitter	0.0225 $[0.0169, 0.0300]$	0.0220 $[0.0164, 0.0294]$	0.0255
PsA	Common grid	0.0085 $[0.0053, 0.0136]$	0.0090 $[0.0057, 0.0142]$	0
PsA	Independent jitter	0.0070 $[0.0042, 0.0117]$	0.0085 $[0.0053, 0.0136]$	0.2042
PsA	Differential jitter	0.0540 $[0.0449, 0.0648]$	0.0570 $[0.0477, 0.0680]$	0.1232

The largest change occurs for PsA when group-1 visits are earlier within nominal windows. LR and GT crossing frequencies rise from 0.0085 and 0.0090 on the common grid to 0.0540 and 0.0570 under differential jitter. The latter estimates have Monte Carlo standard errors 0.0051 and 0.0052, and both Wilson intervals include 0.05. Thus this run does not establish an excess above the nominal bound. It does show that the nominal-bin evidence is sensitive to a visit mechanism that leaves the latent event-time distributions identical. Independent jitter produces wider realized probability ranges but smaller crossing frequencies, illustrating why the association between group and observation timing matters, in addition to the amount of interval variation. All CAV estimates are below 0.05 in the narrower rounding mechanisms examined here; that finding does not justify other visit mechanisms or informative censoring.

The computational supplement contains the scripts, all replication outcomes, full-precision summaries, seed ledger and design record. The seed is $202610220 + b$ for CAV replication b, with 100000 added for PsA; each seed is reused across the three visit mechanisms. Checks on synthetic records compare the completed-interval construction with the original patient-preprocessing functions.